Senior High School

DepED
DEPARTMENT OF EDUCATION

Lesson Exemplar in General Mathematics

QUARTER 3

Unit

7

Lesson Exemplar for General Mathematics
Quarter 3: Unit 7

Development Team

Writers: Aljon S. Bucu PhD
Juan A. Lappay
Ryan B. Bacani

Validators: Jocelyn P. Dela Peña
Samuel A. Quiroz

Language Editor: Dindo John Moreno

Learning Area Specialist: Wilson R. Santiago

Bureau of Learning Delivery
Bureau of Curriculum Development
Bureau of Learning Resources

Every care has been taken to ensure the accuracy of the information provided in this material. For inquiries or feedback, please write or call the Office of the Director of the Bureau of Learning Resources via telephone numbers (02) 8634-1072 and 8631-6922 or by email at blr.od@deped.gov.ph.

Republic of the Philippines

Department of Education

BUREAU OF LEARNING DELIVERY

LESSON EXEMPLAR					
Learning Area:	**GENERAL MATHEMATICS**			Grade Level:	**11**
Semester:	**SECOND**	Quarter:	**THIRD**	Unit:	**7**

I. OBJECTIVES	
Content Standard	*The learners demonstrate knowledge and understanding of basic trigonometry.*
Performance Standard	*By the end of the quarter, the learners are able to apply trigonometry and geometric concepts to estimate measurements and construct three-dimensional objects, understanding their shapes, and volumes for practical uses like product packaging, furniture and construction.*
Learning Competencies	The learners… 1. illustrate the trigonometric ratios in a right-angled triangle using a real-life object or figure. 2. illustrate the area of oblique triangles (i.e Heron's Formula and the rule Area=½ abSinC.) 3. solve practical problems involving: a. right and oblique triangles b. angles of elevation and depression c. the use of bearings in navigation.
II. REFERENCES and MATERIALS	**Textbook and Modules** Mathematics Learner Material, Grade 9 Bacani, J., et. al. *Pre-calculus book Grade 11.* Sunshine Interlinks Publishing House, Inc. 2016 Alferez, Merle. *MSA Trigonometry*, MSA Academic Advancement Institute, 2001. Weimerskirch, M., & Pride, S. (Eds.). (2022). *Trigonometry*. University of Minnesota Board of Regents. https://open.lib.umn.edu/trigonometry/ **Websites**

	https://www.claytonschools.net/cms/lib/MO01000419/Centricity/Domain/204/11_12chapter06.pdf https://coachgriffin.weebly.com/uploads/2/4/6/5/24656764/day_3_-_word_problem_hw_-_key.pdf [Chilimath. (n.d.). *Heron's formula.* https://www.chilimath.com/lessons/geometry-lessons/herons-formula-practice-problems-with-answers/ Cuemath. (n.d.). *Area of Right Triangle.* https://www.cuemath.com/measurement/area-of-right-triangle/ Cuemath. (n.d.). https://www.cuemath.com/measurement/area-of-triangle/ [https://www.jmap.org/Worksheets/G.MG.A.3.HeronsFormula.pdf Mathlobby. (n.d.) https://www.mathlobby.com/post/finding-the-area-of-a-triangle-using-1-2absinc Study.com (n.d.). https://study.com/skill/practice/finding-area-of-a-triangle-using-herons-formula-questions.html **Video Lessons**: Solving Right Triangles | SOH CAH TOA | Trigonometry https://www.youtube.com/watch?v=ljxkJHplMZU Solving Right Triangles - Missing Angle and Side of Right Triangle https://www.youtube.com/watch?v=n4BEwESmzQw Solving Word Problems Involving Right Triangles https://www.youtube.com/watch?v=AyiRdl38aT4 https://www.youtube.com/watch?v=U74R8j4QFMA https://study.com/academy/lesson/video/angles-of-elevation-depression-practice-problems.html https://www.youtube.com/watch?v=uyKvSe6Ltgs https://www.youtube.com/watch?v=ndCmnpoP76M https://www.youtube.com/watch?v=Z1A4fNfSfXM https://www.youtube.com/watch?v=cSKAqfGXOPI **Materials** • Protractor • Calculator • Measuring tape • Clinometer

	• Worksheets
III. CONTENT	**Lesson 7.1: Trigonometric Ratios in a Right-Angled Triangle Using a Real-Life Object or Figure** *Suggested Time Allotment: 2 hours*
IV. OBJECTIVES	At the end of the lesson, the learners are able to: 1. define the six trigonometric ratios (sine, cosine, tangent, cosecant, secant, cotangent) in terms of the sides of a right triangle. 2. illustrate the trigonometric ratios using real-life objects or figures that form right triangles. 3. apply trigonometric ratios to solve practical problems involving right triangles.

V. PROCEDURES	**ACTIVITIES** (Instructions for Learners)	**ANNOTATION** (Instructions for Teachers)
Activating Prior Knowledge	***A.1. Activating Prior Knowledge*** **Activity A.1: Quick Sketch** **Instructions:** 1. On a sheet of paper or in your notebook, draw a right triangle. 2. Choose one acute angle (either of the two smaller angles). 3. Label the sides of the triangle relative to your reference angle: • Hypotenuse – the longest side, opposite the right angle. • Opposite side – the side directly across from the reference angle. • Adjacent side – the side next to the reference angle (but not the hypotenuse). 4. Check that all three sides are properly labeled. 5. Compare your sketch with a partner or groupmate and discuss any differences in labeling. **Reflection Questions**: 1. How did you determine which side was the hypotenuse? 2. What is the difference between the opposite and adjacent sides? 3. Why is it important to identify sides relative to the chosen angle?	**Activity A.1: Quick Sketch** Learners will develop a conceptual understanding of trigonometric side relationships by constructing and labeling their own diagram of a right triangle. **Instructions**: 1. Ask the learners to prepare materials: 2. Present the task and guide learners. 3. Facilitate a brief discussion using the reflection questions.

	4. How do these sides relate to the trigonometric ratios sine, cosine, and tangent?	
	A.2. Establishing the Purpose **Activity A.2:** **Discovering Trigonometric Ratios in Real-Life Triangles** **Instructions:** 1. Form small groups (2–3 members). 2. Study the given real-life scenarios: • A ladder leaning against a wall. • A flagpole casting a shadow. • A kite flying with a string. 3. Draw the right triangle for each scenario and label the sides relative to the chosen angle θ as: • Opposite (O) • Adjacent (A) • Hypotenuse (H) 4. Measure or assign values (your teacher may provide measurements, or you may assume reasonable ones). 5. Calculate the following ratios for each triangle: • Opposite ÷ Hypotenuse (O/H) • Adjacent ÷ Hypotenuse (A/H) • Opposite ÷ Adjacent (O/A) 6. Complete the activity table: [table below] **Reflection Questions:** 1. What do you notice about the ratios when the angle or triangle changes?	**Activity A.2:** **Discovering Trigonometric Ratios in Real-Life Triangle** In this activity, learners will explore how trigonometric ratios (sine, cosine, and tangent) apply real-life situations. **Instructions:** 1. Ask the learners to prepare the necessary materials. 2. Present the task and guide the learners in accomplishing it. 3. Facilitate a brief discussion using the reflection questions. 4. Explain that the activity serves as an application of the lesson in real-life situations. 5. Emphasize that the skills learned in this lesson will be used in the performance task. 6. Present the performance task to the learners.

Scenario	Opposite Side	Adjacent Side	Hypotenuse	O/H	A/H	O/A
Ladder						
Flagpole						
Kite						

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<td>2. Which real-life situation gave the largest tangent ratio? Why?
3. How are the sine, cosine, and tangent ratios related to the sides of the triangle?
4. How can you use these ratios to solve real-world measurement problems?</td>
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<td>B. Instituting New Knowledge</td>
<td>B.1. Presenting Examples

Activity B.1:
Measuring and Calculating Ratios in Right Triangles
Materials:
• Ruler or tape measure
• Protractor
• Right triangle cut-outs (paper, cardboard, or drawn on the board) of different sizes
• Calculator
• Activity sheet
Instructions:
1. Form groups of 3–4.
• Each group will be given at least two right triangles of different sizes.
2. Identify the parts of the triangle:
• Choose one acute angle θ in the triangle.
• Label the sides relative to θ as Opposite, Adjacent, and Hypotenuse.
3. Measure the sides:
• Use a ruler/tape measure to find the lengths of the three sides of the triangle.
• Record your measurements in the activity sheet.
• Calculate the ratios using a calculator:
4. Compute:
a. Opposite ÷ Hypotenuse
b. Adjacent ÷ Hypotenuse
c. Opposite ÷ Adjacent
5. Check consistency:</td>
<td>Activity B.1:
Measuring and Calculating Ratios in Right Triangles
Learners will develop a conceptual understanding of trigonometric side relationships by constructing and labeling their own diagram of a right triangle.
Instructions:
1. Ask the learners to prepare the materials.
2. Present the activity and guide learners in performing the task. Apply any of the following strategies:
Strategy 1:
Hands-on Measurement and Discovery Activity
Learners actively explore the relationship between the sides and angles of right triangles by measuring and computing trigonometric ratios. This promotes conceptual understanding of sine, cosine, and tangent as consistent ratios in similar triangles.
Strategy 2: Discovery Approach
Learners are guided to discover the concept of trigonometric ratios (sine, cosine, and tangent) through measurement, observation, and analysis of right triangles. Instead of being told the formulas, they infer the consistent relationships between side ratios and angles through hands-on exploration and discussion

3. Call a representative from each group to present their output and explain their observations.
4. Wrap up the activity by facilitating the reflection questions.</td>
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- Use a protractor to measure angle θ.
- Compare your calculated ratios

6. Discuss:
   - Are the values or measurement almost equal?
   - Why is there a slight difference in their measurements?

Reflection Questions:
1. What did you notice about the ratios of sides for triangles that have the same acute angle?
2. How do these ratios relate to the trigonometric functions sine, cosine, and tangent?
3. Why do small differences in measurement occur when calculating ratios?
4. How does this activity help you understand trigonometric ratios better?

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Expected answer for the reflection questions:
1. The ratios of the sides are almost the same even if the triangles have different sizes. This means that for a given acute angle, the side ratios remain constant.
2. Each ratio corresponds to a specific trigonometric function:
   - Sine (sin θ) = Opposite ÷ Hypotenuse
   - Cosine (cos θ) = Adjacent ÷ Hypotenuse
   - Tangent (tan θ) = Opposite ÷ Adjacent*
   These functions describe the consistent relationship between the sides of a right triangle and its angles.
3. Slight differences happen due to measurement errors, rounding off, or inaccuracies when using rulers or protractors. These small variations are normal in practical measurement.
4. It shows that trigonometric ratios are based on consistent relationships between angles and sides, regardless of the triangle's size. Measuring and calculating them made it easier to see how sine, cosine, and tangent are formed and used in solving triangles.

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B.2. Discussing the Concept

Activity B.2:
Re-building the Concept of Six Trigonometric Ratios

Instructions:
To summarize and establish what you have learned from the previous activities, listen and participate to the discussions.

In a right triangle, we can define six trigonometric ratios. Consider the right triangle ABC below. In this triangle we let θ represent ∠B. Then the leg denoted by a is the

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Activity B.2:
Re-building the Concept of Six Trigonometric Ratios

Instructions:
In discussing the concept of trigonometric functions, you can do the following:

1. Setting the Context
- Begin by reminding learners of the triangles they measured and the ratios they computed. Say: "Earlier, you worked with opposite/hypotenuse,

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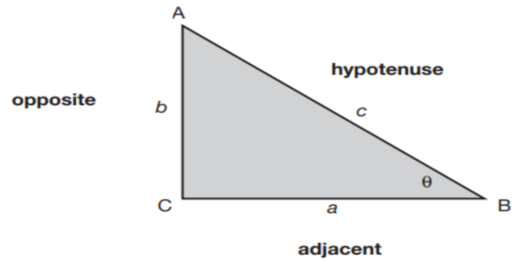
side adjacent to θ, and the leg denoted by b is the side opposite to θ.



We will use the convention that angles are symbolized by capital letters, while the side opposite each angle will carry the same letter symbol, in lowercase.

$$sin\ \ of\ \theta = sin\ \ \theta = \frac{opposite}{hypotenuse}$$

$$cos\ \ of\ \theta = \theta = \frac{adjacent}{hypotenuse}$$

$$tan\ \ of\ \theta = tan\ \ \theta = \frac{opposite}{adjacent}$$

$$\csc\ of\ \theta = csc\ \theta = \frac{hypotenuse}{opposite}$$

$$sec\ of\ \theta = sec\ \theta = \frac{hypotenuse}{adjacent}$$

$$\cot\ of\ \theta = \ \ cot\ \theta = \frac{adjacent}{opposite}$$

SOH-CAH-TOA is a mnemonic used for remembering the equations.

Notice that the three new ratios at the right are reciprocals of the ratios on the left. Applying algebra shows the connection between these functions.

$$csc\ \ \theta = \frac{hypotenuse}{opposite} = \frac{1}{\frac{opposite}{hypotenuse}} = \frac{1}{sin\ \ \theta}$$

adjacent/hypotenuse, and opposite/adjacent. These are just three of six possible ratios in a right triangle."

- Emphasize the naming convention: Angles = capital letters (A, B, C), sides opposite them = lowercase (a, b, c). This builds mathematical discipline in notation.

2. Introducing the First Three Ratios (Primary)

- Write sin θ, cos θ, tan θ clearly with their definitions.
- Use SOH-CAH-TOA as a mnemonic. Say: "Think: Sine = Opposite over Hypotenuse (SOH), Cosine = Adjacent over Hypotenuse (CAH), Tangent = Opposite over Adjacent (TOA)."
- Reinforce by pointing to a triangle drawing on the board and labeling sides relative to θ.

3. Extending to the Other Three Ratios (Reciprocals)

- Introduce csc θ, sec θ, cot θ as reciprocals.
- Stress that they are not new ratios but simply the inverse forms of sin, cos, and tan.
- Use algebra step-by-step to show how csc θ = 1/sin θ, sec θ = 1/cos θ, cot θ = 1/tan θ.

4. Highlighting Connections

- Point out patterns:
  - ➢ "Every trigonometric function has a reciprocal."
  - ➢ "When sin theta is small, csc theta is large, and vice versa."
- This helps learners understand why calculators sometimes show very large values for reciprocal functions.

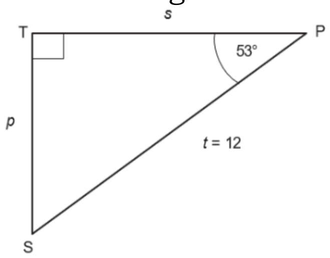
5. Common Misconceptions to Address

$$sec\ \theta = \frac{hypotenuse}{adjacent} = \frac{1}{\frac{adajacent}{hypotenuse}} = \frac{1}{cos\ \ \theta}$$

$$cot\ \theta\ = \frac{adajacent}{opposite} = \frac{1}{\frac{opposite}{opposite}} = \frac{1}{tan\ \theta}$$

Showing a formula for the Missing Parts of a Right Triangle

Example 1: Determine the equation or formula to find a missing part of the triangle.



a. Solve s in the figure 1 above.
Solution: ∠P is an acute angle, t is the hypotenuse, *s* is the side adjacent to ∠P. Use CAH, that is

$$cos\ \ \theta = \frac{adajacent}{hypotenuse}$$

$$cos\ \ P = \frac{s}{t}$$

$$cos\ \ 53° = \ \frac{s}{t} = \frac{s}{12}$$

$$s = 12\,cos\ \ 53°$$

By using scientific calculator, key in 12 cos 53°

s = 7.2218

b. Solve for p in the figure above.
Solution: ∠P is an acute angle, t is the hypotenuse, s is the side adjacent to ∠P.
Use SOH, that is

- Learners often confuse opposite vs. adjacent. Remind them that these depend on the chosen angle.
- Some may think csc θ, sec θ, cot θ are completely different functions. Clarify that they are just reciprocals of the first three.
- Learners might forget that the hypotenuse never changes—it's always opposite the right angle.

6. Teaching Tips

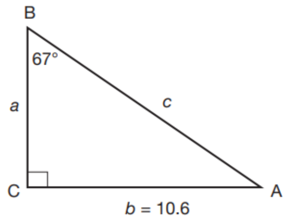
- Visual Aid: Keep a labeled triangle (ABC) on the board throughout the discussion.
- Memory Aid: Use SOH-CAH-TOA chant as a class activity.
- Reinforcement: After presenting the reciprocals, ask learners to pair each function with its reciprocal partner (sin ↔ csc, cos ↔ sec, tan ↔ cot).
- Extension: Challenge learners to compute both sin θ and csc θ for an angle to confirm they are reciprocals.

7. Closure

- Summarize by saying:
  "We now have six trigonometric ratios. Three are primary (sin θ, cos θ, tan θ), and three are their reciprocals (csc, sec, cot). These ratios let us connect angles and sides of any right triangle—and they're the foundation for solving real-world problems."

Note:

1. *Teach the learners in using their scientific calculators.*
2. *Provide additional examples if necessary.*

	$$sin\ \theta = \frac{opposite}{hypotenuse}$$ $$P = \frac{p}{t}$$ $$sin\ 53° = \frac{p}{t} = \frac{p}{12}$$ $$s = 12\, sin\ 53°$$ By using scientific calculator, key in 12 sin 53° s = 9.5836 c. Solve for a in the figure below:  Solution: ∠B is an acute angle, b is the opposite side, and a is the side adjacent to ∠B. Use TOA, that is $$\theta = \frac{opposite}{adjacent} = \frac{b}{a}$$ $$B = \frac{10.6}{a}$$ $$tan\ 67\ ° = \frac{10.6}{a}$$ $$a = \frac{10.6}{tan\ 67°}$$ By using scientific calculator, key in 10.6 divided by tan 67° a = 4.4994 $a\ \approx 4.5$ (a is approximately 4.5) d. Solve for c in figure above.	

Solution: ∠B is an acute angle, b is the opposite side and c is the hypotenuse of the given acute angle.
Use SOH, that is

$$sin\ \theta = \frac{opposite}{hypotenuse}$$

$$B = \frac{b}{c}$$

$$sin\ 67° = \frac{10.6}{c}$$

$$c\ sin\ 67° = 10.6$$

$$c\ \ sin\ 67° = \ 10.6$$

$$c = \frac{10.6}{sin\ 67°}$$

By using scientific calculator, key in 10.6 divided by sin 67°

$$c = 11.5154$$

$c \approx 11.52$ (c is approximately 11. 52)

Example 2: Using the Calculator to Find Trigonometric Ratios

As presented in the previous sections, the values of the trigonometric ratios for any angle are constant, regardless of the length of the sides. These values can also be found using a calculator. Also, you can use a calculator to find an angle when you are given a trigonometric ratio.

a. Finding a ratio given the angle

Example 3:
To find the value of sin 38°, ensure that your calculator is operating in degrees.

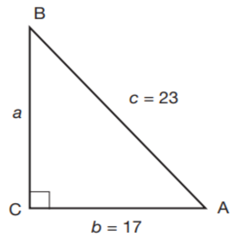
Solution:
Press sin 38 = 0.615661475
The calculator should give sin 38° = 0.616, correct to three decimal places.

	b. Finding an angle given the ratio In finding the size of the angle to the nearest minute, given the value of the trigonometric ratio, just follow the steps in the examples below. *Example 4*: sin θ = 0.725, find θ to the nearest minute *Solution*: Press 2ndF sin 0.725 = 46.46884783 To convert this to degrees/minutes/seconds mode, Press 2ndF D°M'S The calculator gives you 46° 28' (nearest minute) c. Degrees and minutes So far, all angles have been in whole degrees. However, one degree can be divided equally into 60 minutes. Further, one minute can be divided equally into 60 seconds. Angle measurement can also be expressed in degree/minutes form. We can use a calculator to convert a degree measure from decimal form to degree, minute, and second form. *Example 5*: Write 54.46° in degree and minutes, giving an answer correct to the nearest minute. *Solution*: Press 54.46° 2ndF D°M'S The calculator gives 54°27'36", or 54°28'. (nearest minute) In solving right triangles, it means finding the remaining parts. The following are the conditions for the given parts of right triangles:	

Example 6: Triangle BCA is right-angled at C. If c = 23 and b = 17, find ∠A, ∠B and a. Express your answers up to two decimal places.

The is solving a right triangle given the measure of the two parts: the length of the hypotenuse and the length of one leg.

Solution:
Sketch the figure:



a. Side b is the adjacent side of ∠A; c is the hypotenuse of right triangle BCA. Use CAH, that is

$$\cos \cos \theta = \frac{adjacent}{hypotenuse}$$

$$\cos\ A = \frac{b}{c}$$

$$\cos\ A = \frac{17}{23}$$

$$\cos\ A = 0.7391$$

Use scientific calculator to find an angle whose cosine value is 0.7391. Using a scientific calculator, A = 42.340°

b. Since in part (a), it was already found that ∠A = 42.34°, then

$$\angle B = 90° - 42.34°$$

$$\angle B = 47.66°.$$

c. Using the Pythagorean theorem:

$$a^2 + b^2 = c^2$$

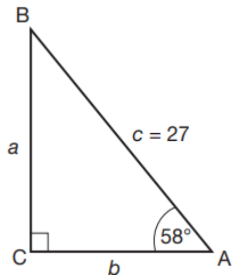
$$a^2 + (17)^2 = (23)^2$$

$$a^2 + 289 = 529$$

$$a^2 = 529 - 289$$

$$a^2 = 240$$

	$a = \sqrt{240}$ $a = 15.49$	
	B.3. Developing Mastery **Activity B.3.** **Practicing Your Skills in Trigonometric Ratios** **Instructions:** Perform the indicated tasks. Write your answer on a sheet of paper. A. Use your calculator to find the value of the following, correct to two decimal places. 1. cos 85° 3. cos 65° 5. tan 63 ° 2. sin 7° 4. tan 35° B. Find the size of the angle θ (to the nearest degree) where θ is acute. 1. $\sin\theta = 0.529$ 4. $\cos\theta = 0.401$ 2. $\cos\theta = 0.493$ 5. $\tan\theta = 2.5$ 3. $\tan\theta = 1.98$ C. Solve the following problems. 1. Triangle BCA is right-angled at C if c = 27 and ∠A = 58°, find ∠B, b, and a. 2. Solve the remaining sides and angles of the right triangle ACB and right angled at C. a. a = 5cm c = 8 cm b. A = 54^0 b = 3 cm **Reflection Questions:** 1. What do you notice about the values of sine, cosine, and tangent as the angle increases? 2. How can the calculator help verify your manual trigonometric computations? 3. Why is it important to round answers consistently in trigonometric calculations	**Activity B.3.** **Practicing Your Skills in Trigonometric Ratios** This activity will help learners to better understand evaluating trigonometric ratios. This can be done by as individual, pair, or group. After accomplishing the activity allow the learners to present their work. Wrap-up the activity by asking the reflection questions. **Key to Corrections** A. Item | Expression | Answer 1. | cos 85° | 0.08 2. | sin 7° | 0.12 3. | cos 65° | 0.42 4. | tan 35° | 0.70 5. | tan 63° | 1.96 B. Item | Given | Answer 1. | sin θ = 0.529 | 32° 2. | cos θ = 0.493 | 60° 3. | tan θ = 1.98 | 63° 4. | cos θ = 0.401 | 66° 5. | tan θ = 2.5 | 68° C. Problem 1.

	4. How do trigonometric ratios help you find missing sides and angles in a right triangle? 5. When is it more convenient to use sine, cosine, or tangent? 6. What is the importance of correctly labeling the sides relative to the given angle? 7. How can this method be applied to real-life measurement problems (e.g., height of a tree or building)?	Analyze the given information.  • The triangle is right-angled at C, so ∠C = 90°. • ∠A = 58° and c = 27 (the hypotenuse). Find the remaining acute angle. • To find B, since B and ∠A are complementary angles, then ∠B + ∠A = 90° ∠B = 90° – 58° ∠B = 32° Calculate the length of the sides. • To find b, since b is the adjacent side of ∠A and c is the hypotenuse of right ΔBCA, then use CAH. $cos\ \theta = \frac{adjacent}{hypotenuse}$ $cos\ \ A = \frac{b}{c}$ $cos\ \ 58° = \frac{a}{27}$ $b = 27\ cos\ 58°$ $b = 27(0.5299)$ $b = 14.31$ • To find a, since a is the opposite side of ∠A and c is the hypotenuse of right ΔBCA, then use SOH. $\theta = \frac{opposite}{hypotenuse}$ $\theta = \frac{b}{c}$ $cos\ \ 58° = \frac{a}{27}$ $a = 27\ sin\ sin\ 58°$ $b = 27(0.8480)$ $b = 22.9$ State the results: $B = 32, \quad a = 22.9\ , \quad b = 14.31$

Problem 2: Solving the Remaining Sides and Angles of a Right Triangle

Given:

$$a = 5\ cm, c = 8\ cm$$
$$A = 54°, b = 3\ cm$$

Step 1: Find b using the Pythagorean Theorem

$$c^2 = a^2 + b^2$$
$$8^2 = 5^2 + b^2$$
$$64 = 25 + b^2$$
$$b^2 = 39$$
$$b = \sqrt{39} = 6.24\ (approximately)$$

Step 2: Find angle A

$$Sin\ A = \frac{opposite}{hypotenuse}$$
$$Sin\ A = \frac{a}{c} = \frac{5}{8} = 0.625$$

$A = arcsin\ arcsin\ (0.625)$ or $A = sin^{-1}(0.0625)$

$$A = 36.68°$$

Step 3: Find angle B

$$B = 90° - A = 90° - 36.68° = 51.32°$$

Final Answers:

b = 6.24 cm, B = 51.32°, $C = 90$

Problem 4. Item 2 Given: A = 54°, b = 3 cm, C = 90°

Procedures:

We need to find a, c, and B

Step 1: Find for angle B

$$B = 90° - A = 90° - 54° = 36°$$

Step 2: Use tangent to find a.

$$tan\ tan\ A = \frac{opposite}{hypotenuse}$$
$$tan\ tan\ A = \frac{a}{b}$$
$$a = b\ tan\ tan\ A$$

		$a = 3\ tan\ tan\ 54°$ $tan\ tan\ 54° = 1.3764$ $a = 3(1.3764)$ $a = 4.13\ cm$ *Step 3*: Find the hypotenuse c $sin\ sin\ A = \frac{opposite}{hypotenuse}$ $sin\ sin\ A = \frac{a}{c}$ $c = \frac{a}{sin\ sin\ A}$ $c = \frac{4.13}{sin\ sin\ 54°}$ $sin\ sin\ 54° = 0.8090$ $c = \frac{4.13}{0.8090}$ $c = 5.10\ cm$ Final answers: $a = 4.13\ cm,\ c = 5.10\ cm,\ B = 36°$
C. Demonstrating Knowledge and Skills	***C.1. Finding Practical Application*** **Activity C.1:** **Estimating the Height Using Trigonometric Ratios** **Instructions:** 1. Form a group of 5 members. 2. Prepare the materials: • Protractor (or smartphone with angle-measuring app) • Measuring tape • String or stick (to help sight the angle) • Calculator 3. Stand a known distance (e.g., 10 meters) from the base of the tree or flagpole. 4. Use the protractor to measure the angle of elevation from your eye level to the top of the tree. 5. Record your eye-level height (e.g., 1.6 meters).	**Activity C.1:** **Estimating the Height Using Trigonometric Ratios** In this activity, learners will apply the trigonometric ratios to measure the height of a real object using indirect measurement. This hands-on activity connects trigonometric concepts with real-world observation and measurement. To facilitate, follow the suggested steps provided. 1. Present the activity. 2. Ask learners to form groups. 3. Distribute the task sheet. 4. Ask learners to prepare the materials. 5. Guide learners in performing the task. 6. Facilitate the presentation to be followed by reflection questions:

	6. Use appropriate trigonometric ratio, to find the height of the tree. 7. Answer the following questions: • Which trigonometric ratio did you use to solve this problem, and why? • How does the angle of elevation affect the height of the kite? • What real-life factors could make your calculation slightly different from the actual height? • In what other real-life situations can you use the same trigonometric principle?	
	C.2. Making Generalization **Activity C.2: What Have I Discovered** **Instructions:** Answer the following questions to summarize what you have learned from the lesson. Write your answers neatly and completely. 1. In your own words, how would you define trigonometric ratios? 2. Why do we use sine, cosine, and tangent in solving right triangles? 3. How can trigonometric ratios be applied in real-life situations? Give one example. 4. Complete the statement below: "I realized that trigonometric ratios are important because ___________________________________."	**Activity C.2: What Have I Discovered** This activity allows learners to reflect on and summarize what they have learned about trigonometric ratios and their applications. It promotes conceptual understanding and encourages learners to express their learning in their own words. By answering open-ended questions, learners summarize key ideas, recognize the purpose of sine, cosine, and tangent in solving right triangles, and identify practical applications of trigonometry in everyday life (such as measuring heights, distances, or angles). This reflective generalization fosters critical thinking, self-awareness, and retention of concepts, preparing learners for higher mathematical applications. **Note to Teacher:** Encourage learners to share their reflections aloud after writing. Use this activity to assess their understanding of both the *conceptual meaning* and *practical applications* of trigonometric ratios. Connect

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<td>their answers to the next topic or performance task for continuity of learning.</td>
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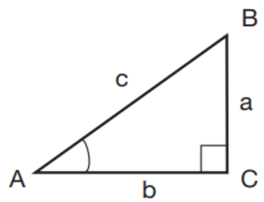
C.3. Evaluating Learning

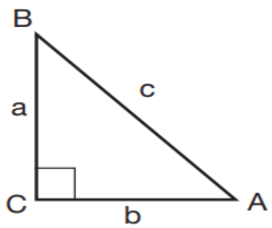
Activity C.3:
Checking Understanding on Trigonometric Ratios

A. Using the figure below, write the expressions that give the required unknown value.



1. If A = 15° and c = 37, find a.
2. If A = 76° and a = 13, find b.
3. If A = 49° and a = 10, find c.
4. If a = 21.2 and A = 71°, find b.
5. If a = 13 and B = 16°, find c.
6. If A = 19° and a = 11, find c.
7. If c = 16 and a = 7, find b.
8. If b = 10 and c = 20, find a.
9. If a = 7 and b = 12, find A.
10. If a = 8 and c = 12, find B.

B. Use the given figure to solve the remaining parts of right triangle ACB.



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Activity C.3:
Checking Understanding on Trigonometric Ratios

At this point, you will assess learners' understanding and application of trigonometric ratios in solving right triangles and real-life problems. To facilitate this activity, follow the instructions below:

1. Introduce the activity by explaining that learners will work independently to apply trigonometric ratios in solving right triangles and related real-life problems.
2. Ensure that learners understand the task directions and have access to all required materials, including calculators and worksheets.
3. Remind learners to identify the reference angle and label the triangle sides correctly as opposite, adjacent, and hypotenuse.
4. Instruct learners to choose and apply the appropriate trigonometric ratio or the Pythagorean theorem based on the given information.
5. Remind them to perform step-by-step solutions and to check their calculator settings (degree mode and rounding).
6. Encourage learners to visualize or sketch each problem to aid in identifying sides and angles.
7. Direct learners to verify the reasonableness of their answers, ensuring that the hypotenuse is the longest side and that angle measures are consistent.
8. Remind them to include correct units (e.g., meters, degrees) in their final answers.
9. Allow learners to explore alternative methods or verify results using another trigonometric ratio.

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	1. b = 17 cm and c = 23 cm 2. c = 16 cm and a = 7 cm 3. b = 10 cm and c = 20 cm 4. b = 6 cm and c = 13 cm 5. c = 13 cm and a = 12 cm C. Solve the following problems: 1. A kite string is 20 meters long and makes an angle of 40° with the ground. How high is the kite above the ground? 2. A ladder 8 meters long leans against a wall making an angle of 65° with the ground. How high up the wall does the ladder reach? 3. A wheelchair ramp is 5 meters long and rises 1.25 meters high. Find the angle of inclination of the ramp with the ground. **Reflection Questions:** 1. Have you completed all the tasks correctly? 2. Which items did you find easy, and which did you find challenging? 3. What common errors did you make? How will you address your misconceptions?	10. Facilitate the activity by monitoring silently, ensuring focus, and collecting outputs for checking and discussion afterward. 14. Synthesize the activity by asking the reflection questions. **Key to Corrections** Part A 1. 9.58 2. 3.37 3. 13.23 4. 7.16 5. 13.54 6. 33.70 7. 14.42 8. 17.32 9.. 30.26° 10. 48.19° Part B 1. $a = \sqrt{23^2 - 17^2} = 15.07$ cm, $A = 41.19°$, $B = 48.81°$ 2. $b = \sqrt{16^2 - 7^2} = 14.42$ cm, $A = 25.84°$, $B = 64.16°$ 3. $a = \sqrt{20^2 - 10^2} = 17.32$ cm, $A = 60°$, $B = 30°$ 4. $a = \sqrt{13^2 - 6^2} = 11.55$ cm, $A = 62.73°$, $B = 27.27°$ 5. $b = \sqrt{13^2 - 12^2} = 5.00$ cm, $A = 67.38°$, $B = 22.62°$ Part C 1. h = 12.86 m 2. h = 7.25 m 3. θ = 14.48°
	C.4. Additional Activities **Activity C.4:** **More Problems on Trigonometric Ratios** **Instructions:** Read and solve each problem:	**Activity C.4:** **More Problems on Trigonometric Ratios** These trigonometry activities are designed to help learners apply sine, cosine, and tangent ratios to solve real-life problems involving right triangles. The

	Problem 1: Height of a Tree From a point on the ground, the angle of elevation to the top of a tree is 35°. If you are standing 15 meters from the base of the tree, how tall is the tree? Problem 2: Shadow of a Building A 10-meter-tall lamp post casts a shadow 6 meters long on the ground. Find the angle of elevation of the sun. Problem 3: Ladder Against a Wall A ladder is leaning against a wall making an angle of 60° with the ground. If the ladder is 5 meters long, how high on the wall does it reach? Problem 4: Slope of a Ramp A wheelchair ramp rises 1.2 meters vertically over a 5-meter length. Find the angle of inclination of the ramp with the ground. Problem 5: Kite Flying A kite is flying at the end of a 40-meter string. If the string makes an angle of 50° with the ground, find how high the kite is above the ground. Problem 6: Surveying a Hill A surveyor stands 120 m from the base of a hill. The angle of elevation to the top is 28°, and a tree halfway up the hill makes an angle of 15° with the horizontal. Find the height of the hill. Problem 7: Observation Tower An observation tower stands on flat ground. From a point 50 m away, the angle of elevation to the top is 40°. From another point 30 m closer to the tower, the angle of elevation is 60°. Find the height of the tower. Problem 8: Flagpole and Building A flagpole is mounted on top of a building. From a point 100 m from the base of the building, the angle of elevation to the top of the building is 30°, and the angle	remediation tasks focus on reinforcing foundational skills in identifying sides, selecting the correct ratio, and calculating missing sides or angles, while the enhancement tasks challenge learners to apply trigonometric concepts in more complex, multi-step scenarios. Through these activities, learners develop problem-solving skills, logical reasoning, and the ability to relate trigonometric principles to practical situations. For learners needing remediation, give problems 1–5, and for those needing enhancement, give problems 6–10. **Answer** 1. 10.5 m 2. 59° 3. 4.33 m 4. 13.9° 5. 30.64 m 6. 56 m 7. 57.7 m 8. 8.7 m 9. 29.6 m 10. a. 0.45 km b. 1.11 km

	to the top of the flagpole is 35°. Find the height of the flagpole. Problem 9: River Crossing A person wants to cross a river and sees a point directly across. They measure the angles of elevation to a hill on the opposite bank from two points 40 m apart along their side of the river: 25° from the first point and 35° from the second. Find the height of the hill. Problem 10: Cable Car Ride A cable car travels up a mountain along a straight cable 1.2 km long, making an angle of 22° with the horizontal. If the cable car station is at the base, find: a. the vertical height gained b. the horizontal distance covered	

III. CONTENT	**Lesson 7.2: Area of Oblique Triangles** *Suggested Time Allotment: 2 hours*	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. illustrate an oblique triangle. 2. compute area of oblique triangles using Heron's Formula and the rule $Area = \frac{1}{2}absinC$ 3. solve problems involving the area of oblique triangles.	
V. PROCEDURES	**ACTIVITIES**	**ANNOTATION**
A. Activating Prior Knowledge	**A.1. Activating Prior Knowledge** **Activity A.1: Triangle Recall Challenge** **Instructions:** 1. Look at the different triangles be shown by your teacher. 2. Identify and label the parts of each triangle (sides, base, height, and included angle). 3. Recall the formula for finding the area of a right triangle and write it on your paper. 4. Using the given measurements, compute the area of the right triangle using the formula $A = \frac{1}{2}bh$. 5. Answer the guide question:	**Activity A.1: Triangle Recall Challenge** This activity will determine learners' prior understanding of triangle properties and their existing knowledge of finding the area of triangles, especially right triangles. **Materials:** Paper or worksheet with triangle drawings Ruler and calculator Protractor (optional for angle identification) **Instructions:** 1. Present different types of triangles (right, acute, obtuse, scalene).

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What will you do if the triangle is not a right triangle?
Can you still find its area? Explain your answer.

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2. Ask learners to:

Identify the parts of each triangle (sides, included angle, base, height).
Recall and write the formula for the area of a right triangle.
Compute the area of a given right triangle using $A = \frac{1}{2}bh$.

3. Facilitate a short discussion by asking the guide questions.

Expected Output:
Learners’ responses will reveal whether they can recall the basic area formula for right triangles and understand triangle parts, this will serve as a foundation for introducing the area of oblique triangles using Heron’s and sine formulas.
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A.2 Establishing the Purpose
Activity A.2: The Farmer’s Triangular Field
Instructions:
Read the problem and do the indicated task.

Scenario:
A farmer owns a triangular plot of land that he wants to fence and plant crops in. The plot is not a right triangle, and he knows the lengths of the three sides: 40 meters, 55 meters, and 60 meters.

He needs to figure out how much space he has in the field, but he realizes that the usual area formula for right triangles cannot be used because there is no perpendicular height.

Task:

Draw the triangular field and label the sides with the given measurements.
Discuss with your group:

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Activity A.2: The Farmer’s Triangular Field
This activity will help learners realize that not all real-world triangles are right triangles and that there are ways to find their area even without perpendicular height.

To facilitate this activity, follow the instructions below.

Prepare the Materials: Ensure each learner or group has paper, pencils, and a ruler. Optionally, display a diagram of a triangular field on the board or projector.
Present the Scenario: Read the “Farmer’s Triangular Field” scenario aloud to the class. Emphasize that the triangle is not a right triangle and the farmer wants to know the area.
Guide Learners to Draw: Ask learners to draw the triangle and label the sides with the given measurements (40 m, 55 m, 60 m).
Facilitate Discussion: Encourage learners to discuss in pairs or small groups how the farmer might find the area without knowing the height. Walk around to monitor participation and prompt ideas.

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	• How could the farmer find the area of his land without knowing the height? • What challenges might arise if he tried to use the area formula for right triangles? • Write down your group's ideas and possible solutions.	5. **Ask Reflective Questions:** • Why can't we use the usual right-triangle area formula? • What strategies could we use to find the area of this triangle? 6. **Have Groups Share:** Invite each group to share their thoughts and possible methods. Summarize the discussion, highlighting that special formulas (Heron's or sine formula) are needed for oblique triangles. 7. **Conclude:** Reinforce that this activity connects to real-world applications and prepares learners to learn new methods for finding the area of oblique triangles in the next lesson and then present the performance task.
B. Instituting New Knowledge	***B.1. Presenting Examples*** **Activity B.1: Exploring Triangular Plots** **Instructions:** 1. Study the three triangles to be showed by your teacher. Each triangle has different side lengths and included angles. o Triangle A: sides 8 cm and 10 cm, included angle 90° o Triangle B: sides 8 cm and 10 cm, included angle 60° o Triangle C: sides 8 cm and 10 cm, included angle 30° 2. Using your knowledge of finding the area of a right triangle, compute the area of Triangle A. 3. Observe Triangles B and C. Without using any formula, predict: o Which triangle will have the largest area? o Which will have the smallest? 4. Discuss in pairs or groups what might cause the difference in area among the three triangles even though they have the same sides.	**Activity B.1: Exploring Triangular Plots** The previous activity introduces a real-life situation where learners realize that the usual formula for finding the area of a right triangle cannot be used for an oblique triangle, creating the need to discover a new method. This is followed by the *Exploring Triangular Plots* activity, where learners examine triangles with the same sides but different included angles and observe that the area changes as the angle changes. Through this exploration, learners begin to understand that the area of a triangle depends not only on its sides but also on the included angle, leading them naturally to the concept and derivation of the formula $A = \frac{1}{2}ab\sin C$ for finding the area of oblique triangles. To facilitate this activity, follow the instructions below. 1. Prepare visuals of the three triangles (drawn to scale or printed).

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<td>5. Share your observations with the class.</td>
<td>2. Ask probing questions while learners are comparing the triangles, such as:
• “What do you notice about the included angle and the area?”
• “How does changing the angle affect the size of the triangle?”
3. Guide learners to conclude that the area of a triangle depends not only on the lengths of its sides but also on the size of the included angle.
4. Transition by saying:
“In our next activity, we will learn how to compute the exact area of any triangle like these even if it isn’t a right triangle.”</td>
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<td>B.2. Discussing New Concept

Activity B.2.:
Computing the Area of Non-right Triangles

Instructions:
Listen attentively and participate actively in the discussion.
Definition:
Oblique triangle - An oblique triangle is simply a triangle without a right angle. It may be acute (all angles < 90°) or obtuse (one angle > 90°).

Area of Triangle Formula
The area triangle can be calculated using various formulas. For example, Heron’s formula is used to calculate the triangle’s area, when we know the length of all three sides. Trigonometric functions are also used to find the area of a triangle when we know two sides and the angle formed between them.

The Heron's Formula (SSS)
Finding the Area of a Triangle Using Heron’s Formula
When all three sides of a triangle are known and no angle is given, we can find its area using Heron’s</td>
<td>Activity B.2.:
Computing the Area of Non-right Triangles

In this activity, learners will study how to determine the area of oblique triangles using two different methods: the trigonometric rule and Heron’s formula. Through guided examples and problem-solving tasks, they will further understand how these formulas apply. The activity aims to help learners connect geometric concepts with real-life applications, such as finding the area of triangular fields, roofs, or lots where measuring height directly is not possible. By comparing both methods, learners will develop a deeper understanding of when and how to use each formula effectively.

To facilitate this activity, follow these steps:
1. Begin by connecting the previous activities:
Start by asking learners how they would find the area of a triangle if it is not a right triangle. Encourage them to recall the basic formula for the area of a right triangle and identify what is missing in an oblique one (the height).
2. Introduce the concept:
Explain that even if a triangle has no right angle</td>
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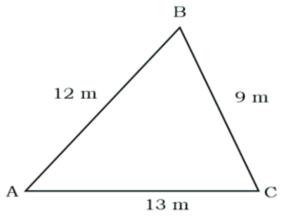
	Formula. This method does not require the height or any angles — only the side lengths. The formula is: $A = \sqrt{s(s-a)(s-b)(s-c)}$ where: • A= area of the triangle • a, b, c= lengths of the three sides • s= semiperimeter of the triangle, calculated as $s = \dfrac{a+b+c}{2}$ **Example 1:** Mang Lito's farm plot has sides measuring 12 meters, 9 meters, and 13 meters. What is the area of his plot? Solutions: 1. Given triangle sides: a = 9m, b = 13 m, and c =12 m, we can illustrate the triangle plot.  2. Calculate semi-perimeter (s): $s = \dfrac{a+b+c}{2}$ $s = \dfrac{9+13+12}{2}$ $s = \dfrac{34}{2}$ $s = 17\,m$ 3. Apply Heron's Formula: $A = \sqrt{s(s-a)(s-b)(s-c)}$ $A = \sqrt{17(17-9)(17-13)(17-12)}$	specifically the oblique triangle, its area can still be found using special formulas derived from trigonometry and geometry. 3. **Present the two methods:** **First Method:** When all three sides are known, introduce **Heron's Formula** $A = \sqrt{s(s-a)(s-b)(s-c)}$,where $s = \dfrac{a+b+c}{2}$ Explain how this formula works even without knowing any angles. **Second Method:** When two sides and their included angle are known, introduce the formula $A = \dfrac{1}{2}ab\sin\,C$ Discuss how the sine function helps find the height indirectly. 4. **Present examples:** *Provide additional examples if necessary.* Apply any of the following strategies: • *Guided Problem Solving* Steps: ➢ Present a step-by-step problem involving an oblique triangle. ➢ Work through the first steps together, asking learners to calculate each stage. ➢ Gradually release responsibility, letting learners complete the rest of the problem individually or in small groups. ➢ Ask learners to present their solutions • *Think-Pair-Share* Steps ➢ Ask learners to individually solve a problem. ➢ Next, have them pair up to share their ideas and compare approaches.

	$A = \sqrt{17(8)(4)(5)}$ $A = \sqrt{2720}$ $A = \sqrt{2720}$ $A \approx 52.15\, m^2$ Therefore, the area of $\triangle ABC$ is about 52.15 square meters. **The Trigonometric Rule** When a triangle is not a right triangle, its height is not easily measured. However, if two sides and the included angle are known, we can find its area using a trigonometric rule. The formula is: $A = \frac{1}{2}ab\sin C$ where: • A= area of the triangle • a and b= lengths of the two sides • C= included angle between sides aand b This rule comes from the idea that the sine of an angle relates the height of the triangle to one of its sides. It works for both acute and obtuse triangles and is very useful in real-life applications such as land measurement, navigation, and construction when the perpendicular height cannot be measured directly. **Using Trigonometric Rule** **Example 2:** Mang Pedro is building a triangular roof beam. He measures two sides: • Side a = 15 meters • Side b = 20 meters The angle between these two sides is 70°. **Question:** Find the area of the triangular roof beam. Solutions:	➢ Finally, invite pairs to share their findings with the whole class. ***Note: Provide more examples if necessary***. 5. **Encourage conceptual understanding:** Ask learners to compare the two methods and identify which situations each is best suited for. Allow them to discuss how these formulas can be applied in real-life scenarios such as land measurement or construction.

	Given: a = 15 m, b = 20 m, C = 70^0 $$A = \frac{1}{2}absinC$$ $$A = \frac{1}{2}(15)(20)sin(70^0)$$ $$A = 150sin(70^0)$$ $$A \approx 140.96\,m^2$$ Therefore, the area of the triangle is approximately 140.96 square meters.	
	B.3. Developing Mastery **Activity B.3:** **Enhancing Skills in Finding the Area of Oblique Triangles** **Instructions:** Solve the following problems individually. Show complete step-by-step solutions. Afterward, pair-check with your seatmate, then volunteer to explain one problem on the board. **Problem 1.** A triangular fish pen in Laguna de Bay has sides 16 m, 18 m, and 25 m. Find its area. **Answer:** A ≈ 143.56 m^2 **Problem 2.** During typhoon season, a barangay is constructing a triangular roof for an evacuation center. The roof beams measure a = 20 m, b = 25 m, with an included angle of 40°. Find the roof's triangular area. **Answer:** A ≈ 160.70 m^2 **Problem 3.** A triangular rice field in Isabela has sides 40 m, 55 m, and 70 m. Find its area. Round to two decimal places. **Answer:** A ≈ 1097.85 m^2	**Activity B.3:** **Enhancing Skills in Finding the Area of Oblique Triangles** This activity is designed to help learners master the application of triangle area formulas particularly the Heron's Formula and ½ absinC in real-world contexts. By solving practical problems involving triangular structures such as fish pens, roofs, fields, and stages, learners will strengthen their understanding of geometry and its relevance to engineering, agriculture, and community planning. To facilitate this activity, choose any of the following strategies. **Scaffolded Practice with Fading Support** **Instructions:** 1. Start Problem 1 together — teacher provides cues (e.g., *"What is the semi-perimeter?"*). 2. For Problem 2, guide only the formula selection, then let learners perform calculations. 3. For Problem 3–5, learners solve independently, teacher only intervenes if needed. 4. Summarize by asking the reflection questions.

	Problem 4. In Cagayan Valley, engineers are reinforcing a flood control dike. A triangular concrete support has two sides measuring 32 m and 27 m, with an included angle of 55°. Find the area of the triangular support. **Answer:** A ≈ 353.87 m² **Problem 5.** The barangay built a triangular stage with sides 10 m, 15 m, and 17 m. Find its area. **Answer:** A ≈ 74.46 m² **Reflection Questions:** 1. Which problem was easiest or most challenging, and why? 2. How did working with your seatmate help you understand the steps better? 3. How did choosing the correct formula affect your solution? 4. How can the skills learned be applied in real-life situations like construction or agriculture?	**Distributed Practice with Immediate Feedback** **Instructions:** 1. Assign Problems 1–5 for independent solving. 2. As learners work, walk around and check for: ● Incorrect formula choice ● Missing units ● Wrong calculator mode (radians vs. degrees) 3. Provide quick corrections: *"Remember, divide by 2 to get s."* or *"Always use the included angle."* 4. Collect 1–2 answers from volunteers after each problem to verify with the whole class. 5. Facilitate the reflection questions. **Error Analysis and Peer Teaching** **Instructions:** 1. Prepare a "common mistake" sample (e.g., forgetting √ in Heron's formula, using the wrong angle). 2. Present the solution on the board and ask: *"Spot the error. What went wrong?"* 3. After discussion, pair learners to peer-check each other's solutions to Problems 3–5. 4. Call volunteers to re-teach the correct solution on the board. 5. Facilitate the reflection questions.
C. Demonstrating Knowledge and Skills	***C.1. Finding Practical Application*** **Activity C.1:** **Applying Triangles in Real-Life Design** **Instructions:** Analyze the scenario and perform the indicated tasks. **Scenario:** Your barangay is planning community improvement projects: a small plaza, a new evacuation center, and	**Activity C.1:** **Applying Triangles in Real-Life Design** This activity engages learners in applying triangle geometry to realistic community projects: a plaza garden, an evacuation center roof, or a rice field. Learners will calculate areas of oblique triangles, estimate required materials, and compute total costs.

	optimizing local rice fields. Many of these projects involve triangular spaces: triangular garden plots in the plaza, triangular roof trusses for typhoon-resistant buildings, and irregular triangular-shaped rice fields.
	
	Your group will act as community planners, engineers, or agricultural advisors to model these triangles, calculate their areas, and make practical decisions, including estimating the amount of materials needed and the cost involved.
	• Plaza Garden Plot:
	    o Each triangular garden plot needs soil, grass, or paving stones.
	    o Assume 1 bag of soil covers 2 m^2 and costs ₱350 per bag.
	    o Calculate how many bags are needed for your triangle and the total cost.
	• Evacuation Center Roof Truss:
	    o Each triangular truss covers a portion of the roof.
	    o Assume 1 sheet of roofing material covers 2 m^2 and costs ₱500 per sheet.
	    o Calculate how many sheets are needed for your truss and the total cost.
	    o Discuss how the triangle's shape affects stability during typhoons.
	• Rice Field Plot:
	    o Farmers need fertilizer for their triangular rice field.
	    o Assume 1 kg of fertilizer covers 10 m^2 and costs ₱50 per kg.
	    o Calculate how much fertilizer is needed and the total cost for your triangle.
		The activity promotes critical thinking, problem-solving, and real-life application of mathematics in community planning, construction, and agriculture
		
		1. **Introduce the Scenario**
		    • Explain the context: the barangay is planning improvements that involve triangular spaces.
		    • Highlight the real-life applications: landscaping, construction, and agriculture.
		    • Emphasize the importance of calculating area accurately to plan materials and budget.
		2. **Form Groups**
		    • Divide learners into groups of 3–5.
		    • Assign each group a triangle scenario from the prepared data table (plaza, roof, or rice field).
		3. **Demonstrate the Process**
		    • Model one example on the board or projector:
		        o Show how to use the formula Area = $\frac{1}{2}ab\sin C$.
		        o Show how to convert the area into material needs and total cost.
		    • Explain how to label sides and angles clearly when modeling triangles.
		4. **Monitor and Support Group Work**
		    • Allow groups to model their triangle either physically (chalk, tape, measuring tape) or digitally (GeoGebra, Desmos, map app).
		    • Check that learners correctly identify sides, angles, and units of measurement.
		    • Circulate to guide calculation of area, material needs, and total cost without giving answers directly.
		5. **Presentation and Sharing**
		    • Each group presents their triangle model, area calculation, materials, and cost analysis.

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Your task is to measure, calculate, and interpret these triangles to make real-life recommendations for your barangay.

Procedures:

1. **Form Your Group**
   - Work in a group of 3–5 members.
   - Your teacher will provide a triangle data card (two sides and the included angle) representing a plaza plot, roof truss, or rice field.
2. **Model the Triangle**
   - **Option A:** Draw the triangle on the ground using chalk, masking tape, or measuring tape**.**
   - **Option B:** Use a simulation tool like GeoGebra, Desmos, or a map app to model the triangle.
   - Label all sides and angles clearly.
3. **Calculate the Area**
   - Use the formula appropriate formula.
   - Record your calculation on the worksheet.
4. **Estimate Materials and Cost**
   - Use the scenario's material coverage and costs to compute:
     - Plaza: Number of soil bags and total cost.
     - Roof: Number of roofing sheets and total cost.
     - Rice Field: Kilograms of fertilizer and total cost.
5. **Present Your Work**
   - Share your triangle model, area calculation, material estimates, and costs with the class.
   - Explain how your calculations help in planning, construction, or agriculture**.**
6. **Reflect**

   Answer these questions**:**
   - What did you learn about applying triangle geometry in real-life situations?

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- Encourage learners to explain their reasoning and how their calculations help in real-life planning.

6. **Reflection and Discussion**
   - Lead a discussion using reflection questions:
     - What did you learn about applying triangle geometry in real-life situations?
     - How can these calculations help in budgeting, planning, or resource allocation?
     - Why is knowing the area of triangular spaces important for efficiency, safety, and cost management?
7. **Closure**
   - Summarize key takeaways: connecting geometry with practical decision-making.
   - Highlight the importance of accuracy in calculations and how math informs real-life planning.

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- How can these calculations help in budgeting, planning, or resource allocation?
- Why is knowing the area of triangular spaces important for efficiency, safety, and cost management?

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C.2. Making Generalization

Activity C.2:
The Triangle Detective – Which Formula Fits?

Instructions:
1. Examine the three **triangle evidence cards** below:
   - **Evidence A:** A triangular rice field has sides 24 m, 30 m, and 18 m. (Only three sides are given.)
   - **Evidence B:** A barangay solar panel has two beams measuring 15 m and 20 m with an included angle of 45°.
   - **Evidence C:** A flood control dike forms a triangle with all three sides given: 10 m, 14 m, and 12 m.
2. For each evidence card:
   - Circle or write the correct formula: **Heron's Formula** or $\frac{1}{2}absinC$.
   - Write a one-sentence justification: *"I chose this because…"*
3. After matching all evidence cards, complete the **Detective's Final Report**:

Detective's Generalization:
- Use Heron's Formula when ______________
- Use $\frac{1}{2}absinC$ when __________________________.

Share your **"Final Report"** with a partner, then be ready to read your rule aloud in class.

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Activity C.2:
The Triangle Detective – Which Formula Fits?

In this activity, learners act as "triangle detectives" to determine which formula is appropriate for finding the area of different triangles. They will analyze evidence cards representing real-life situations (rice fields, solar panels, and flood control dikes), justify their formula choice, and develop a general rule for when to use each formula. This activity strengthens conceptual understanding of triangle area formulas, critical thinking, and reasoning skills.

The instructions below will you the guide to facilitate this activity.

Instructions
1. **Introduce the Activity**
   - Explain the detective theme: learners are tasked with figuring out the "correct formula" for each triangle scenario.
   - Briefly review the two formulas.
2. **Distribute Evidence Cards**
   - Provide each learner with the three evidence cards (Evidence A, B, and C).
   - Ensure each student has a worksheet to record formula choices and justifications.
3. **Individual Analysis**
   - Learners examine each card and decide which formula to use.
   - They must write a one-sentence justification for each choice:

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		• "I chose this because…" • Encourage learners to consider what information is given (sides, angles) before choosing a formula. 4. **Complete Detective's Final Report** • Learners generalize the rule for formula use: • Circulate to check for understanding and provide guidance if a student is confused. 5**. Partner Sharing** • Have learners pair up to share their Final Report. • Encourage discussion and comparison of reasoning. 6**. Reporting** • Invite a few learners to read their general rule aloud. • Summarize correct generalizations on the board. 7**. Closure** • Reinforce the importance of analyzing given information before choosing a formula. • Highlight the practical applications in real-life scenarios (e.g., agriculture, construction, engineering).
	C.3. Evaluating Learning **Activity C.3:** **Assessing Understanding on the Area of Oblique Triangles** **Part I – Concept Recall** 1. Define an oblique triangle and give one real-life example in your community. 2. Explain briefly when to use Heron's Formula and when to use the trigonometric rule $Area = \frac{1}{2} absinC$.	**Activity C.3:** **Assessing Understanding on the Area of Oblique Triangles** This culminating activity allows learners to demonstrate their understanding of computing the area of oblique triangles using both Heron's Formula and the trigonometric rule $A = \frac{1}{2} ab\sin C$. The activity combines concept recall, real-life applications, and

	3. Complete the sentence: "We use Heron's Formula when ________, and we use $Area = \frac{1}{2}absinC$ when ________." **Part II – Real-Life Applications** Solve the following problems completely. Show step-by-step solutions and include units. 1. The DPWH is surveying a triangular lot needed for a road widening project in Quezon City. The lot has sides measuring 35 m, 40 m, and 50 m. Find its area so they know how much land to acquire. 2. After a powerful typhoon in Albay, a family is rebuilding their home with stronger triangular roof supports to withstand future storms. Two support beams measure 18 meters and 22 meters, forming an included angle of 42°. Using trigonometry, calculate the area of the triangular roof support. 3. The city government of Baguio is designing a small triangular park with sides 20 m, 26 m, and 30 m. Find the park's area so they can estimate grass and landscaping costs. 4. A company in Metro Manila is installing a triangular solar-powered billboard. Two sides measure 14 m and 19 m, with an included angle of 37°. Find its area. If each square meter costs ₱1,200 to install, how much is the total cost? **Part III – Reasoning & Analysis** **Read and analyze each problem below and answer the questions.** 1. A student solved for the area of a triangle with sides 20 m, 25 m, and 30 m using $Area = \frac{1}{2}absinC$. Is the formula used correct? Why? If it's incorrect, what formula should have been used?	reasoning tasks to assess both procedural fluency and analytical thinking. In part I learners, review key concepts and differentiate when to use each formula. In part II, learners solve practical problems involving community-related scenarios such as construction, land surveying, and engineering projects, showing complete step-by-step solutions. In part III, learners justify formula choices and evaluate problem-solving approaches, reinforcing conceptual understanding and critical reasoning. Through this activity, learners apply mathematics to realistic situations, interpret results meaningfully, and strengthen their ability to choose appropriate methods for solving problems involving oblique triangles. Each learner should complete this activity on their own. It plays a key role in evaluating how well they understand the lesson. Be sure to check their work right away so you can offer quick and suitable help to those who may need extra support or reinforcement. **Answer: s** Part I 1. An oblique triangle is a triangle that does not contain a right angle. (Answers may vary) 2. A triangular garden plot in the barangay plaza. (Answers may vary) 3. We use Heron's Formula when all three sides are known, and we use Area = ½ ab sin(C) when two sides and the included angle are known. Part II 1. Area ≈ 695.27 m^2

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<td>2. If you only know two sides of a triangle and a non-included angle, can you directly apply Heron's Formula or $Area = \frac{1}{2}absinC$? Justify your answer.</td>
<td>2. Area ≈ 132.49 m²
3. Area ≈ 256.25 m
4. Area ≈ 80.04 m²
Total Cost ≈ ₱96,049.68 (at ₱1,200 per m²)

Part III
1. Incorrect. The formula Area = ½ ab sinC is only valid when you know two sides and the included angle between them. In this case, the student was given three side lengths, so the correct formula to use is Heron's Formula, which is specifically designed for triangles with all three sides known.
2. No, you cannot directly apply either formula. Heron's Formula requires all three sides. Area = ½ ab sinC requires two sides and the included angle (the angle between the two known sides)</td>
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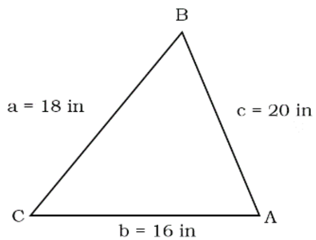
<td>C.4. Additional Activities

Activity C.4:
More Exercises on the Area of Oblique Triangles

Remediation
Instructions:
Solve the following problems. Show your complete solutions.

1. Determine the area of triangle ΔABC shown in the figure. Round your answer to the nearest hundredths.

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<td>Activity C.4:
More Exercises on the Area of Oblique Triangles

This extra activity can be assigned as homework. Provide remediation for learners who haven't fully mastered the lesson and offer enrichment tasks to those who are ready to advance their learning further.

Key to Correction
Remediation:
1. 137.74 cm²
2. 139.63 in²
3. 35.66 square units
4. 25.98 cm²
5. 256.16 m²

Enrichment
1.
$$Area = \frac{1}{2}(10)sin\,(55^0) \approx 65(0.8192) \approx 53.25\,m^2$$
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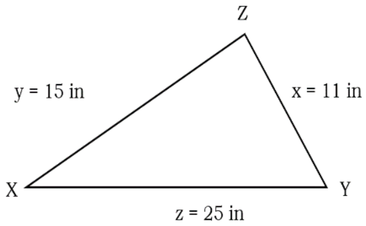
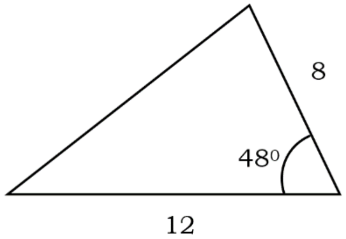
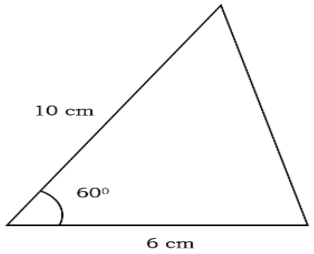
2. Calculate the area bounded by ΔXYZ in the given figure below. Round your answer to the nearest hundredths.



3.What is the area of the region bounded by the triangle on the right? Round your answer to the nearest hundredths.



4. Find the area of the triangle below.



5. Mang Edong is measuring a triangular section of land for irrigation. Two sides of the plot are 30 m and 25 m, with an included angle of 45°. Find the area of the triangular section.

Enrichment
Instructions:
Solve the following problems that follow.

Problem 1: A triangle has sides b=10 m and c=13 m, with included angle A=55^0.
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$Area = \frac{1}{2}(8)(11)\ sin\ sin\ (70^0)\ \approx 44(0.9397) \approx 41.35\ ft.^2$
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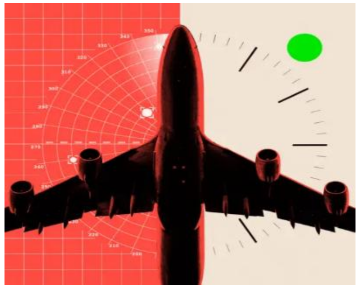
	- Sketch the triangle: - Draw side b = 10 m. - At one end, use a protractor to draw a 55° angle. - From that angle, draw side c = 13 m. - Connect the endpoints to complete the triangle. - Label the triangle with sides and angle. - Calculate the area. Problem 2: A triangle has sides a=8 ft. and c=11 ft., with included angle 70^{0}. - Sketch the triangle**:** - Draw side a = 8 m. - At one end, use a protractor to draw a 70° angle. - From that angle, draw side c = 11 m. - Connect the endpoints to complete the triangle. - Label the triangle with sides and angle. - Calculate the area:	

III. CONTENT	**Lesson 7.3: Shapes Beyond the Surface: Application of Right and Oblique Triangles** *Suggested Time Allotment: 2 hours*	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. use the trigonometric ratios to solve problems involving angle of elevation and depression. 2. apply the laws of sines and cosines in solving problems involving oblique triangles. 3. solve situational problems involving bearings in navigations.	
V. PROCEDURES	**ACTIVITIES**	**ANNOTATION**
A. Activating Prior Knowledge	***A.1. Activating Prior Knowledge*** **Activity A.1** **Geometry on the Move: Choosing the Shortest Path** **Instructions:** Analyze the scenario below, then make a logical decision. **Scenario**:	**Activity A.1** **Geometry on the Move: Choosing the Shortest Path** This activity simply leads the learners to recalling the idea of Pythagorean Theorem as applied in Right Triangles. To facilitate the activity, follow the strategy below. **Instructions:** 1. Begin the discussion by sharing this prompt question.

	Lance wants to walk from Point A to Point C in a park. Point B is directly between A and C, forming a right triangle where: • Point A is 300 meters west of Point B, • Point C is 400 meters north of Point B. **Guide Questions**: 1. Make a clear illustration of the situation by using pen and paper or Desmos/Geogebra. 2. Should Lance walk from A to B and then from B to C, or should he walk directly from A to C? 3. Which path is shorter, and by how much? 4. What mathematical concept or principle that guides you in making decisions?	"Have you ever needed to find the shortest path between two places? How did you decide which way to go?' 2. Let the class work in pairs and brainstorm about the given situation. 3. Present the title of the activity ***"Geometry on the Move: Choosing the Shortest Path"*** so that the learners will foresee what the activity is all about. 4. Inform them that the task requires practical and logical analysis such as the use of math facts. 5. Tell them to either visualize the problem using manila paper, Geogebra or Desmos. 6. Remind them to consider the guide questions as indicated before they make their best decision. 7. Provide feedback from each presentation or allow other learners to give their inputs.
	A.2. Establishing the Purpose of the Lesson **Activity A.2:** **Navigating with Angles and Discovering the Math Behind Exploration** **Instructions:** The teacher will show picture, reflect on the picture and answer the questions related to the pictures  **Reflection Questions**: 1. What enables pilots and boat captains to navigate vast skies and open seas with such precision, without getting lost?	**Activity A.2:** **Navigating with Angles and Discovering the Math Behind Exploration** This engaging activity introduces learners to the real-world connection between navigation and mathematics. Through guided discussion and critical thinking prompts, learners reflect on travel experiences and the role of tools like maps, compasses, and GPS. By synthesizing their ideas, learners realize that navigation depends on accurate angle measurements just like in trigonometry. The activity concludes by linking this concept to ancient Filipino navigators who skillfully used stars, wind, and waves to journey across seas, highlighting the timeless importance of mathematical thinking in exploration. The suggested instructions below may apply to facilitate this activity. **Instructions:**

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2. What do you think helps pilots and captains stay on course during their journey?
3. How do they know where they are going, especially when they're surrounded by sky or sea?
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1. Begin by asking:
• Have you ever traveled by plane or ship?
• Would you like to be a pilot or a ship captain someday?
• Let learners share their experiences or aspirations. This sets a personal and imaginative tone.
2. Prompt Critical Thinking
• Encourage learners to mention tools like GPS, compass, maps, radar, and even stars.
3. Synthesize Ideas
• Write down all student responses on the board or chart paper.
• Highlight any mention of angles, directions, or measurements.
• Guide the discussion toward the idea that navigation relies on precise angle measures to stay on track whether it's the angle of elevation to a star or the bearing of a ship.
4. Connect to Mathematics
In this activity, we'll explore how ancient Filipino navigators used the stars, wind, and waves to travel across seas without any modern technology. They applied principles we now study in trigonometry, geometry, and astronomy.
5. Present the performance task.
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<td>B. Instituting New Knowledge</td>
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B.1. Presenting Examples

Activity C.1:
Differentiating Right and Oblique Triangle

Instructions:
Study the two figures below, compare them based on their characteristics. Present your answer using a Venn Diagram.
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Activity B.1:
Differentiating Right and Oblique Triangle

In this activity, learners analyze two given triangles by observing their characteristics and identifying both differences and similarities. Using a Venn Diagram, they will organize their observations clearly to show how the figures compare. This activity encourages critical thinking, visual analysis, and understanding of geometric properties while helping students develop skills in classification and representation.
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Figure 1

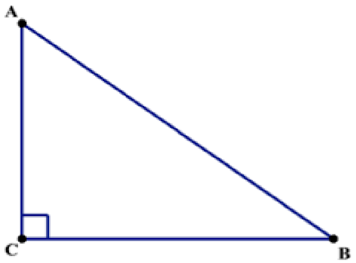
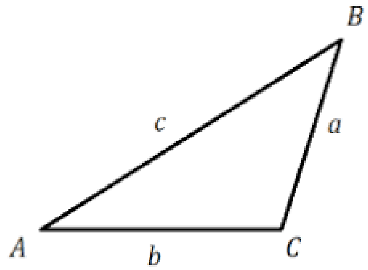


Figure 2



Guide Questions.

1. What makes the two triangles different?
2. How about their similarities?
3. How can you present or organize your responses using a Venn Diagram?

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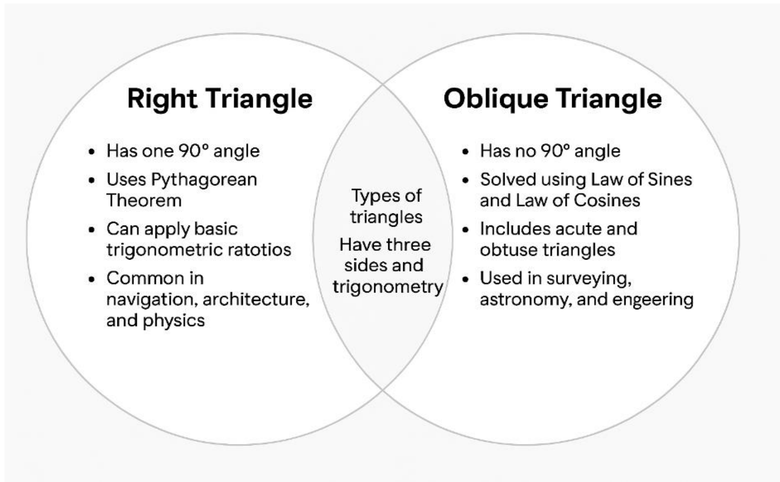
Instructions:

1. Present these figures as a starting concept leading to the main topic.
2. Tell the learners to formulate their thoughts by looking into the similarities and differences of the figures.
3. Inform them to organize their thoughts using a graphic organizer, say Venn Diagram.
4. Let them discuss the work either by pair or individually.
5. Assess the performance of the learners using a rubric.
6. Summarize the outputs by giving the wrap-up ideas.
7. Tell the class that the task brings them to concept of oblique triangle.

Possible Answer



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B.2. Discussing the Concept

Activity B.2:
Up and Down, We Go Exploring Angles in Real Life

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Activity B.2:
Up and Down, We Go Exploring Angles in Real Life

This part of the lesson is critical, as learners will develop important conceptual understanding. To facilitate the

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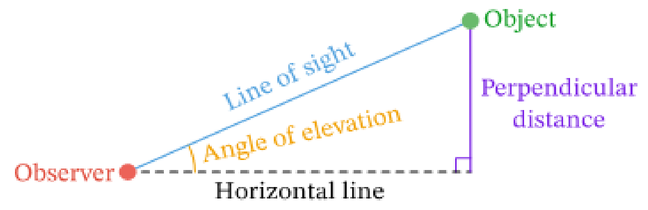
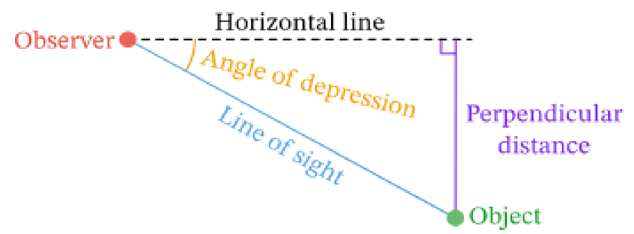
Instructions:
The following concepts will be covered in this lesson. Explore the activities carefully to develop a clear understanding of each concept.

Since you now knew the trigonometric ratios, it means that you are ready to apply them in solving problems that involve elevation and depression. Let us differentiate them through illustration.

Right Triangle for an Angle of Elevation



Right Triangle for an Angle of Depression



Source:
https://share.google/images/IkOHLVJFBWlaLi3nt

When you observe an object positioned above your eye level, the angle formed between your horizontal line of sight and the upward direction to the object is called the ***angle of elevation.***

discussion, you may apply any of the following strategies.

Provide additional examples if necessary.

Strategy 1: **Technology-Assisted Discussion**
Instructions:
1. Display on the screen each individual task to be assigned to every group.
2. Maximize the use of technology such as graphing calculators, Geogebra, etc.
3. Each group will plug-in the given facts in the problem
4. Gather their outputs and post them.
5. Let each representative to explain briefly about the output.

Strategy 2**: Guided Instruction**
Instructions
1. Present a Realistic Scenario
2. Define the Problem Clearly
3. Gather and Analyze Relevant Information
4. Plan a Solution Strategy
5. Execute the Plan
6. Evaluate and Interpret the Solution
7. Reflect on Learning

Strategy 3: **Scaffolded Learning**
Instructions
1. Introduce the Problem
2. Break the Problem into Smaller Parts
3. Model the First Few Steps
4. Provide Guided Practice
5. Encourage Independent Problem Solving
6. Check and Discuss Solutions

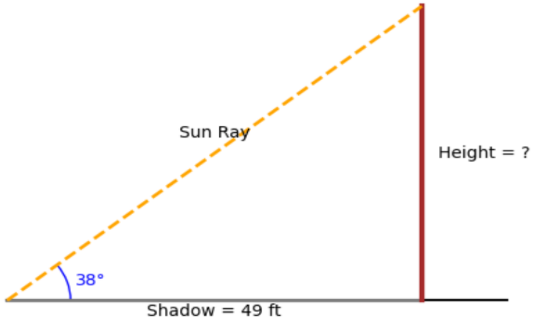
	Conversely, when the object is located below your eye level, the angle between your horizontal line of sight and the downward direction to the object is known as the ***angle of depression.***	

Example 1:

Sunlight hits a flagpole, casting a shadow that stretches 49 feet along the ground. At that moment, the sun's rays form a 38° angle with the ground, how tall is the flagpole?

Solution.

Show the solution by illustrating the problem first:



The given side is the adjacent side with respect to the angle of elevation 38^0. To determine the trigonometric ratio that corresponds to the problem, it is useful to apply a mnemonic device called $SOH - CAH - TOA$ that stands for:

$$SOH\text{- } \sin\theta = \frac{opposite}{hypotenuse}$$

$$CAH\text{- } \cos\theta = \frac{adjacent}{hypotenuse}$$

$$TOA\text{- } \tan\theta = \frac{opposite}{adjacent}$$

Thus, we use the trigonometric ratio on tangent.

$$\tan\theta = \frac{opposite}{adjacent}$$

Let x be the height of the flagpole (opposite side), adjacent side of 49 ft. with $\theta = 38^0$. Hence, we have

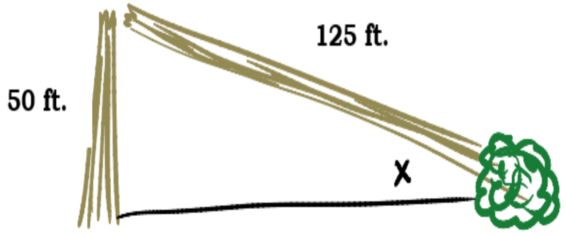
$$\tan 38 = \frac{x}{49}$$

Solving for x,

$$x = 49\,(\tan 38) = 38.28\,ft.$$

This means that the height of the flagpole is 38.28 ft long.

Example 2.
A tree is struck by lightning and snaps off 50 feet above the ground. The top part of the tree, 125 feet long, rests with the tip on the ground while the broken end rests on the top of the stump. What angle does the top part of the tree make with the ground?



Solutions
To solve for the angle x, note that sides given are the opposite side and the hypotenuse. So, we apply

$$\sin\theta = \frac{opposite}{hypotenuse}$$

$$\sin x = \frac{50}{125}$$

$$\sin x = 0.4$$

The problem asks for the angle, so we use the **Inverse Function.** Know that the domain of the inverse function of sine is $-1 \leq x \leq 1$ and its range $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$.
Since, 0.4 belongs to the domain, hence, the range exists.

$$x = sin^{-1}(0.4)$$

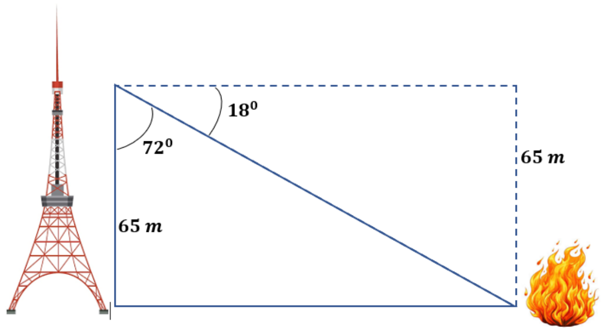
$$x = 23.58^0.$$

The angle that the top part of the tree make with the ground is 23.58^0.

Note: *Be sure that your calculator is in DEGREE Measure or if in RADIAN measure the value of x should be multiplied to* $\frac{180}{\pi}$ *or* 57.3^o.

Example 3.
A ranger stationed in a fire tower that stands 65 meters above the ground spots a fire below. The ranger's line of sight to the fire forms an angle of depression of 18°. How far is the fire from the base of the tower horizontally?

Solutions:



This problem can be solved in two methods – one that uses 18^0 and 72^0 as reference angles.

Method 1. For $\theta = 72^0$, we apply

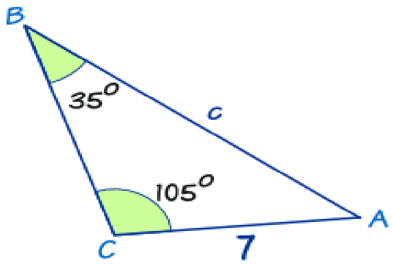
$$tan\ 72 = \frac{x}{65} = x = 65(tan\ 72) \sim 200\ m.$$

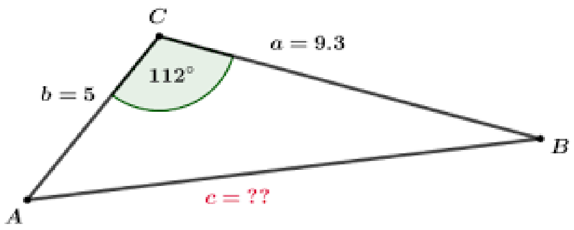
Method 2. For $\theta = 18^0$, we have

$$tan\ 18 = \frac{65}{x} = x = \frac{65}{tan\ 18} \sim 200\ m.$$

The distance of the fire from the base of the tower horizontally is approximately 200 m.

Consider the triangles as shown below. How do you solve for the unknown sides and angles? This question unfolds the discussion about some laws such law of sine and cosine. In order to solve any problems involving oblique triangles, let us know more about the triangle congruences.





By Congruence Axioms, two triangles are congruent if their corresponding sides and angles are equal.

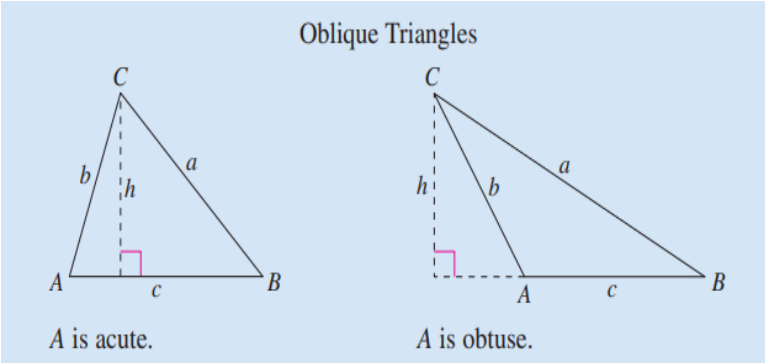
These congruence axioms include Side-Angle-Side (SAS), Angle-Side-Angle (ASA), Side-Side-Side (SSS) which are very useful in understanding the laws of sine and cosine in an oblique triangle.

An oblique triangle refers to a triangle that does not contain 90^0 angle measure in any of the three angles. It is very important to know that a triangle can be solved if at least one side and any other measures are given.

Law of Sine in Oblique Triangle

If ABC is a triangle with sides a, b *and* c, then

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

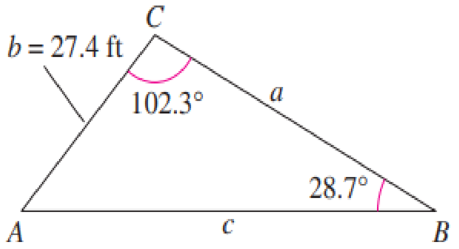


We can rewrite this form in terms of its reciprocal.

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Example 4.

Find the remaining sides and angle of the triangle below.



Solutions:
Let us determine the third angle A

$$\angle A = 180^0 - (102.3^0 + 28.7^0) = 49^0$$

Now, we can find the rest of the sides by using the Law of sine

$$\frac{sin\ A}{a} = \frac{sin\ B}{b} = \frac{sin\ C}{c}$$

To get side c, we have

$$\frac{sin\ 28.7}{27.4} = \frac{sin\ 102.3}{c}$$

$$c\ (sin\ 28.7) = 27.4\ (sin\ 102.3)$$

$$c = \frac{27.4\ (sin\ 102.3)}{sin\ in\ 28.7} = 55.75\ ft.$$

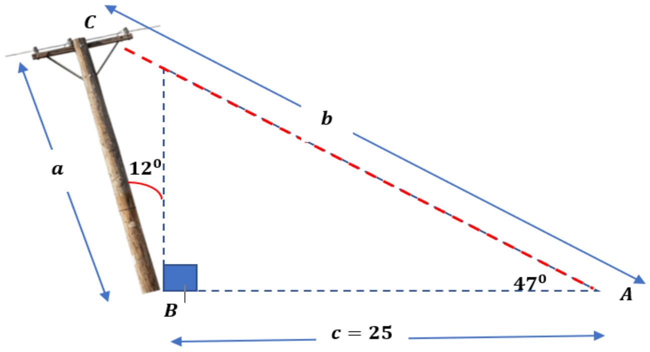
Finally, side a gives

$$\frac{sin\ 28.7}{27.4} = \frac{sin\ 49}{a}$$

$$a\ (sin\ 28.7) = 27.4\ (sin\ 49)$$

$$a = \frac{27.4\ (sin\ \ 49)}{sin\ 28.7} = 43.06\ ft.$$

Example 5.
A pole tilts *toward* the sun at 12^0 from the vertical, and it casts a shadow of 25 ft. The angle of elevation from the tip of the shadow to the top of the pole is 47^0. Find the height of the pole.



Solutions.
We start with determining the angle measures of $A, B \ and \ C$.

$$\angle A = 47^0, \angle B = 90^0 + 12^0 = 102^0$$

To find for $\angle C = 180^0 - (47^0 + 102^0) = 31^0$.

Using the Law of Sine, we get

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin 47}{a} = \frac{\sin 31}{25}$$

$$a \sin 31 \ = 25 \sin 47$$

$$a = 25 \sin 47 \left(\frac{1}{\sin 31}\right) \ = 35.5 \ ft.$$

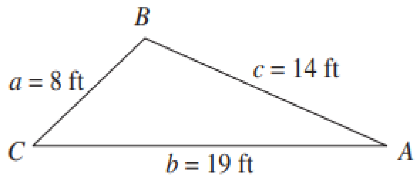
The height of the pole is 35.5 ft.

LAW OF COSINE IN OBLIQUE TRIANGLE

If the three sides of the triangle are SSS and SAS (two sides and their included angle), Law of sine cannot be applied. In such cases, *Law of Cosine* will be used.

Standard Form	*Alternative Form*
$a^2 = b^2 + c^2 - 2bc \cos A$	$\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$
$b^2 = a^2 + c^2 - 2ac \cos B$	$\cos B = \dfrac{a^2 + c^2 - b^2}{2ac}$
$c^2 = a^2 + b^2 - 2ab \cos C$	$\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$

Example 6.
Find the measure of the three angles of the triangle below.



Solutions:
In this case, we can start the solution by finding the angle opposite the longest side which is $\angle B$.

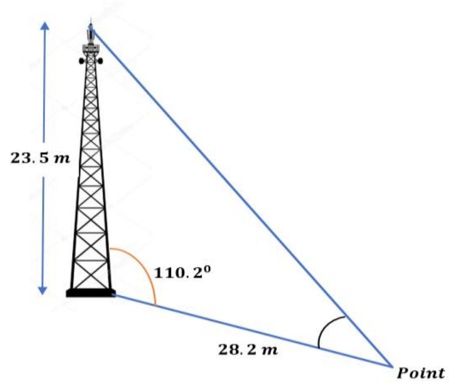
By Law of cosine, we have

$$cos\ B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$cos\ B = \frac{8^2 + 14^2 - 19^2}{2(8)(14)} = -0.4508$$

Using the inverse function of cosine,

$$cos\ B = -0.4508$$
$$\angle B = arccos(-0.4508)$$

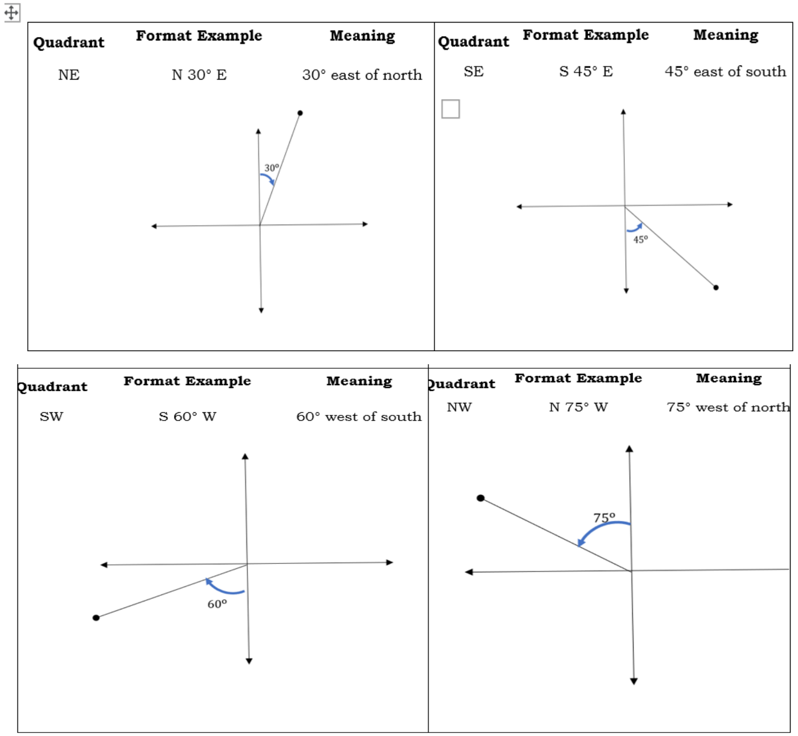
	$\angle B = 116.8^0$ To find angles A and B, Law of sine may be applied for a simpler solution. $$\frac{\sin A}{a} = \frac{\sin B}{b}$$ $$\frac{\sin A}{8} = \frac{\sin 116.8}{19}$$ $$\sin A = \frac{8\,(\sin 116.8)}{19} = 0.3758$$ $$\angle A = \arcsin\,(0.3758) \;= 22.08^0$$ $$\angle C = 180 - (116.8 + 22.08) = 41.12^0.$$ **Example 7** A 23.5 – meter tall tower forms an angle of 110.2° with an inclined road. A point is located 28.2 meters down the road from the base of the tower. What is the angle subtended by the tower at that point?  **Solutions:** We first calculate the third side of the triangle (the line of sight from the top of the tower to the point down the road), say side c whose opposite angle C is 110.2^0. By Cosine law,	

	$$c^2 = a^2 + b^2 - 2ab\,(\cos C)$$ $$c^2 = 23.5^2 + 28.2^2 - 2(23.5)(28.2)\cos 110.2$$ $$c = \sqrt{1805.15} = 42.49\ m$$ To determine the angle subtended by the tower at that point, we use Sine Law. Let x be the point 28.2 m from the base of the tower. $$\frac{\sin x}{23.5} = \frac{\sin 110.2}{42.49}$$ $$\sin x = \frac{23.5 \sin 110.2}{42.49} = 0.5191$$ By inverse function, we get $$x = \arcsin(0.5191) = 31.27^0$$ Therefore, the angle subtended by the tower at that point is 31.27^0. **Bearings in Navigations** *Steps to Determine and Name Navigation Direction* Using Rotation from Due North **Step 1: Understand the Reference Line** • The North-South line is your reference. • Due North is considered 0°. • Rotation is measured clockwise from Due North. **Step 2: Measure the Bearing Angle** • Use a protractor or compass rose to measure the angle of rotation clockwise from Due North to the direction of travel. • This angle is called the bearing. **Step 3: Identify the Quadrant** • Navigation bearings are typically expressed using quadrant notation or true bearings. • Quadrant Bearings Format Bearings are named using the starting cardinal direction, the angle, and the ending cardinal direction.	

- The angle is always less than 90°.

Quadrant	Format Example	Meaning
NE	N 30° E	30° east of north
SE	S 45° E	45° east of south
SW	S 60° W	60° west of south
NW	N 75° W	75° west of north

Sample illustrations:



True Bearings Format:
Measured clockwise from Due North (0° to 360°).
Examples:

- Due East = 90°
- Due South = 180°
- Due West = 270°

- N 30° E = 30°
- S 45° W = 225°

Step 4: Name the Direction

Depending on the format used:

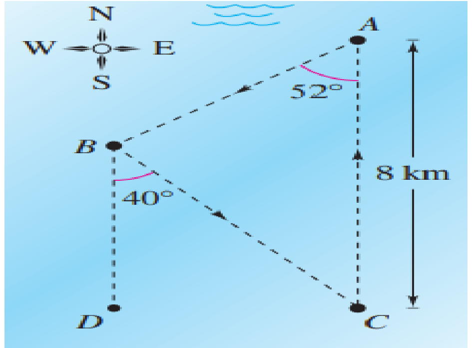
- **Quadrant Bearing**: Use the cardinal directions and angle (e.g., **N 40° E**).
- **True Bearing**: Use the angle only (e.g., **40°** or **225°**).

Step 5: Apply in Navigation Problems

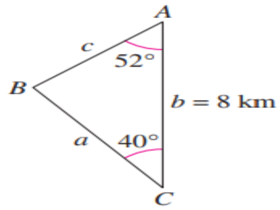
- Use trigonometry (sine, cosine, tangent) to calculate distances and directions based on bearings.
- Bearings help determine the direction of movement in triangle-based navigation problems.

Example 8.

The course for a boat race starts at point A and proceeds in the direction $S\,52^0\,W$ to point B, then in the direction $S\,40^0\,E$ to point C, and finally back to A, as shown below. Point C lies 8 kilometers directly south of point A. Approximate the total distance of the race course.



From the illustration, $BD \parallel AC$ and is cut by a transversal BC. This means that $\angle DBC \cong \angle BCA = 40^0$, hence, we have



Consequently, $\angle B = 180 - (40 + 52) = 88^0$. By Law of Sines, we can determine side a

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\frac{\sin 52}{a} = \frac{\sin 88}{8}$$

$$a = \frac{8\sin 52}{\sin 88} = 6.31\ km$$

$$c = \frac{8\sin 40}{\sin 88} = 5.15\ km$$

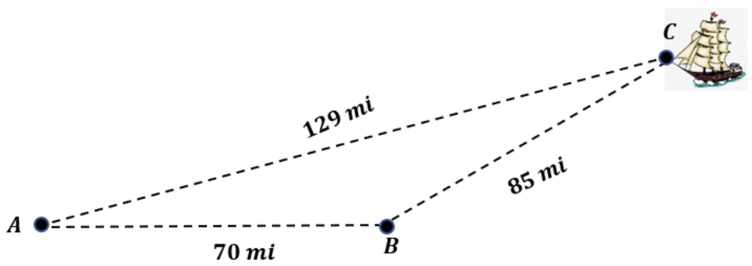
To solve for the total length of the course, simply get the sum of the three sides, thus, we get

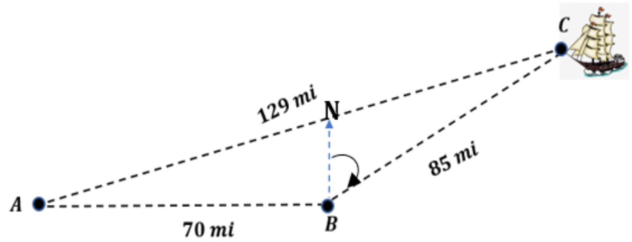
$$Total\ length = 8 + 6.31 + 5.15 = 19.46\ km.$$

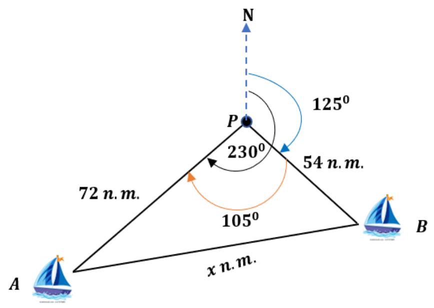
Example 9.

A ship first travels 70 miles directly east, then changes direction and heads northward. After covering 85 miles in this new direction, the ship is found to be 129 miles away from its original starting point.

Describe the bearing from point B to point C.



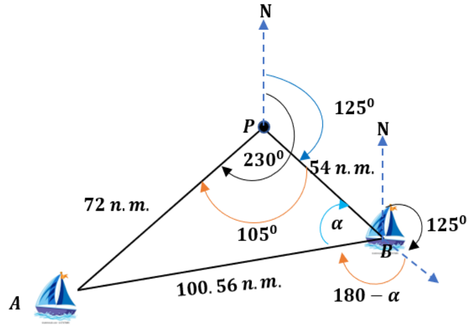
	Solutions: Since three sides are given, then you can apply the Law of Cosine, this gives you $$\cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{85^2 + 70^2 - 129^2}{2(85)(70)} = -0.3795$$ So, $B = arc\ cos(-0.3795) = 112.30^0$. Note: *Bearings are measured clockwise from North.*  Hence, the bearing taken from due north from point B to point C is $112.30^0 - 90^0 = 22.30^0$ **or** $N\ 22.30^0\ E$. **Example 10.** Two ships leave the same port at the same time. One ship sails on a course of 125^0 at 18 knots while the other sails on a course of 230^0 at 24 knots. Find after 3 hours (a) the distance between the ships and (b) the bearing from the first ship to the second. **Solutions:** Let P be the port. After 3 hours, the first ship is at point B and the second ship is at point A. The first ship travels at 18 knots, making a distance of 54 nautical miles from point P to B. The second ship travels at 24 knots, making a distance of 72 nautical miles from point P to A. The angle at P in the triangle is $230^0 - 125^0 = 105^0$	



Suppose the distance between the two ships after 3 hours is x nautical miles. By law of cosine, you get

$$x^2 = 72^2 + 54^2 - 2(72)(54)\cos 105$$
$$x = 100.56\ n.m$$

a. The bearing from the first ship to the second ship after 3 hours is the bearing from B to A.



To find the bearing, first we find angle α and the angle at B in the triangle. Use the law of sine.

$$\frac{\sin\alpha}{72} = \frac{\sin 105}{100.56}$$

$$\sin\alpha = \frac{72\sin 105}{100.56}$$

$$\sin\alpha = 0.6916$$

$$\alpha = arcsin\,(0.6916) = 44^0$$

Now, we notice that the bearing B to A is $125^0 + (180 - \alpha) = 125^0 + (180^0 - 44^0) = 261^0$.

B.3. Developing Mastery

Activity B.3: Delving Deeper on Triangles

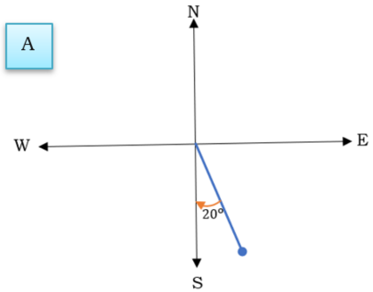
Instructions:
Perform the indicated tasks.

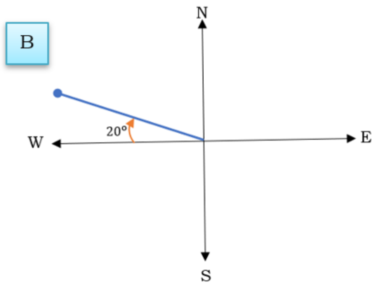
Exercise 1.
Match the measure of the given bearings with the appropriate graph below. Write the letter of your answer on the space before each number.

Given:

_____ 1. 20^o
_____ 2. $S\ 70^o\ W$
_____ 3. 160^o
_____ 4. $S\ 20^o\ W$
_____ 5. $N\ 70^o\ W$
_____ 6. 340^o
_____ 7. 110^o
_____ 8. 270^o
_____ 9. 180^o
_____ 10. $N\ 70^o\ E$.

Graphs:





Activity B.3: Delving Deeper on Triangles

This activity will provide learners with an opportunity to deepen their understanding of the lesson. To make it more interactive, you may apply any of the following strategies.

Strategy 1. **Think-Pair-Solve-Share**
Instructions:
1. Give each student a set of problems.
2. Ask them to solve one problem alone (Think).
3. Then, pair them with a seatmate to discuss and solve the next one together (Pair + Solve).
4. Finally, invite volunteers or randomly chosen pairs to share their answers and explain their reasoning to the class (Share).
5. You may use visual aids like posters or digital slides to show each problem.
6. Facilitate the following reflection questions:
   - Which types of problems in this activity did you find easiest or most challenging, and why?
   - How did understanding triangle properties and trigonometric ratios help you solve real-life navigation and distance problems?
   - How can the skills and concepts practiced in this activity be applied in everyday situations, such as travel, construction, or aviation?

Strategy 2. **Presenter-Reactor Feud**
Instructions:
1. Use a small group to facilitate the activity.
2. Provide each group a copy of the task. From each group, there shall be a presenter and a reactor.

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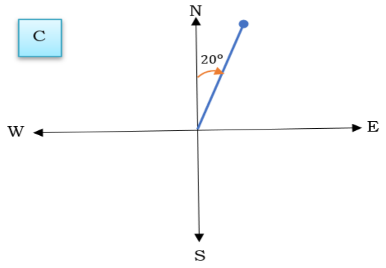

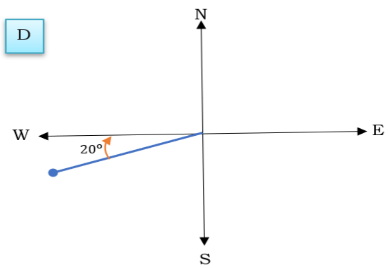

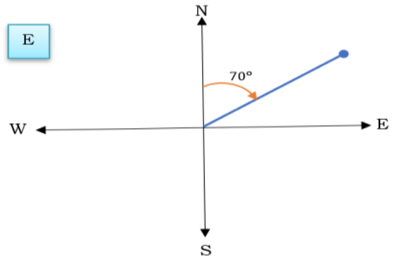

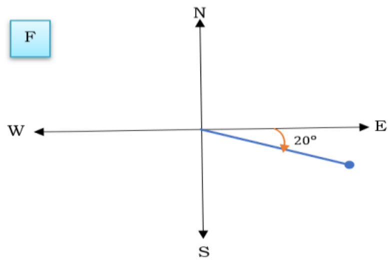

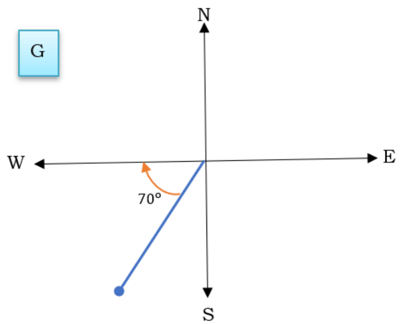

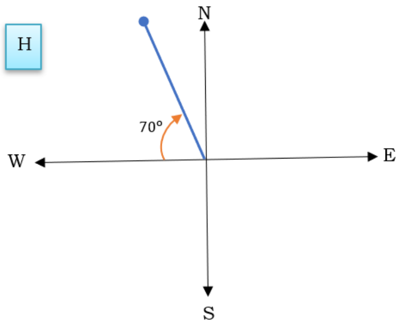

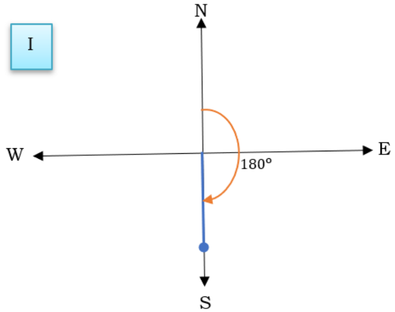

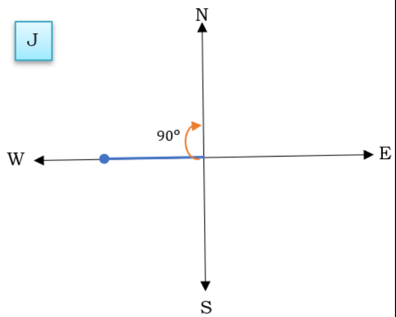

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3. Assign a presenter who shall present the output of the group and shall discuss and explain clearly the final output.
4. Instruct the remaining groups or non-presenting groups to designate reactors who shall task to provide feedback with regard to the presenter's discussion and point out areas of improvement.
5. Assign a moderator to facilitate the presenter-reactor discussion.
6. Conduct FGDs among the groups to summarize the discussions.
7. Facilitate the following reflection questions:
   - Which types of problems in this activity did you find easiest or most challenging, and why?
   - How did understanding triangle properties and trigonometric ratios help you solve real-life navigation and distance problems?
   - How can the skills and concepts practiced in this activity be applied in everyday situations, such as travel, construction, or aviation?

Key to Corrections

Exercise 1.

1. C
2. D
3. A
4. G
5. B
6. H
7. F
8. J
9. I
10. E

Exercise 2.

1. hypotenuse = $\sqrt{2}$units

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Exercise 2.

Solve the problems below.

1. The legs of an isosceles right triangle are 1 unit long. Find the length of the longest side in simplest radical form.
2. Use part (a) to find the exact value of each of the following. a. $tan\ 45^0$ b. $sin\ 45^0$ c. $cos\ 45^0$
3. An airplane is at an elevation of 35, 000 ft when it begins its approach to an airport, its angle of descent is 6^0. What is the distance between the airport and the point on the ground directly below the ground?
4. A student looks out of a second-story school window and sees the top of the school flagpole at an angle of elevation of 22^0. The student is 18 ft above the ground and 50 ft from the flagpole. Find the height of the flagpole.
5. Observers at points A and B, 30 km apart, saw an airplane at angles of elevation of $40^0 and\ 75^0$, respectively. How far is the plane from each observer?
6. A ship proceeds on a course of 300^0 for 2 hours at a speed of 15 knots (1 knot=1 nautical mile per hour). Then it changes course to 230^0, continuing at 15 knots for 3 hours. At that time, how far is the ship from its starting point?

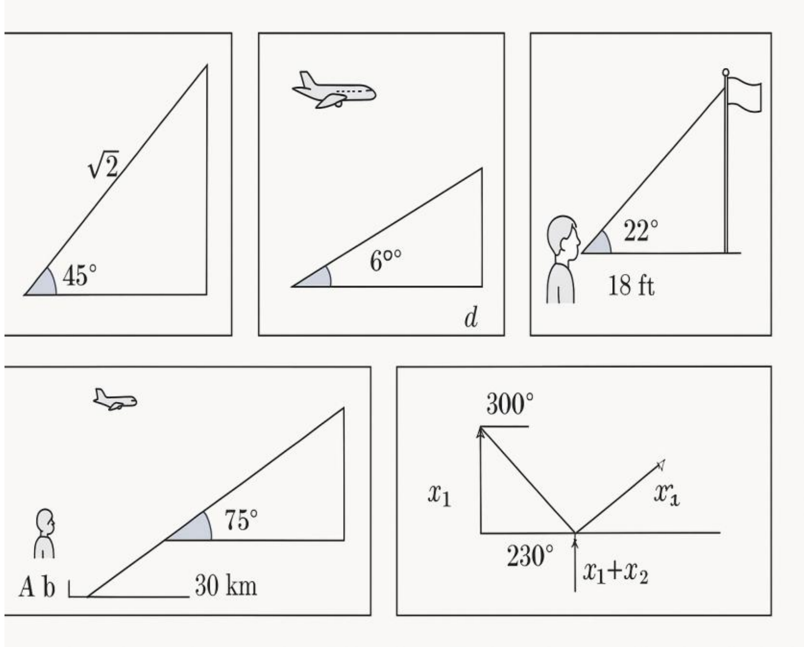
2. $\tan 45^\circ = \frac{\text{opposite}}{\text{adjacent}} = \frac{1}{1} = 1$

$$\sin\ 45^\circ = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\cos\ 45^\circ = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

3. distance ≈ 333,050 ft or 63.1 miles
4. height of flagpole ≈ 38.2 ft
5. plane is 24.48 km from A, 5.52 km from B, height ≈ 20.53 km
6. ship is ≈ 62.1 nautical miles from starting point

Sample Illustrations:



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<td>C. Demonstrating Knowledge and Skills</td>
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C.1. Finding Practical Application

Activity C.1:
Mission Navigation: Find Your Way!

Instructions:
Analyze the scenario below and perform the indicated task.

Scenario:
You are part of a rescue team tasked with locating a missing vessel based on bearing instructions. You must calculate positions, distances, and bearings using trigonometry and vector analysis.

Materials:

Compass rose diagrams
Graph paper or coordinate grids
Protractors and rulers
Calculators
Scenario cards with bearing-based navigation tasks

Task:
A rescue boat leaves port on a bearing of 45° and travels 100 km. It then receives a signal to change course to 120° and travels another 80 km.

Goal: Determine the boat’s final position and its bearing from the port.

Learners Roles:
Navigator: Calculates bearings and distances
Cartographer: Draws the path on a grid
Reporter: Explains the solution and reasoning

Reflection Questions:

What strategies did your team use to calculate bearings and distances, and how effective were they?

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Activity C.1:
Mission Navigation: Find Your

In this activity, learners apply trigonometry and vector analysis to a real-world navigation scenario. Working as a rescue team, they calculate distances, bearings, and positions to locate a vessel, integrating mathematical reasoning with practical problem-solving.

The activity encourages collaboration, critical thinking, and the application of geometry and trigonometry in authentic contexts.

Instructions:

Present the scenario and explain the roles of Navigator, Cartographer, and Reporter.
Provide the necessary materials, including compass rose diagrams, graph paper, protractors, rulers, calculators, and scenario cards.
Guide learners to analyze the bearing instructions and distances, encouraging discussion and reasoning among group members.
Monitor group work, offering prompts or hints if learners struggle with calculations or plotting vectors.
Facilitate the sharing of solutions, ensuring that each group explains both their process and final answers.
Highlight connections between the activity and real-life navigation to reinforce the practical application of mathematical concepts.
To end the session, ask the reflection questions.

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	2. How did collaborating in different roles (Navigator, Cartographer, Reporter) help in solving the navigation problem? 3. How can the mathematical concepts you applied in this activity be used in real-life navigation or other practical situations?	
	C.2. Making Generalization **Activity C.2:** **Learning Ticket: My Trip Through the Lesson** **Instructions:** Complete each statement below. Write your answer on a separate sheet of paper. 1 **Best Stop** The part of the lesson that made the most sense was... 2 **Bumpy Road** The part of the lesson i have trouble with was... 3 **Funny Moment** The part of the activity that i enjoyed the most was.. **One Thing I'll Remember** If I would explain this topic to a friend, I would say	**Activity C.2:** **Learning Ticket: My Trip Through the Lesson** This activity allows learners to consolidate their understanding by reflecting on key concepts from the lesson. By completing guided statements, students actively summarize what they have learned, identify important takeaways, and generalize based on the activities and discussions. This encourages critical thinking and helps teachers assess learners' comprehension. **Instructions**: 1. Provide each learner with a copy of the guided statements or have them write responses on a separate sheet of paper. 2. Explain that the purpose of the activity is to reflect on the lesson and summarize key concepts. 3. Encourage learners to think critically and express their understanding in their own words. 4. Monitor and provide support if learners have difficulty articulating their responses. 5. Collect and review the completed learning tickets to assess conceptual understanding and identify areas that may need further clarification.
	C.3. Evaluating Learning **Activity C.3: Assessing Your Understanding** **Instructions:** Read and solve each problem.	**Activity C.3: Assessing Your Understanding** Each learner should complete this activity on their own. It plays a key role in evaluating how well they understand the lesson. Be sure to check their work right

1. Find the measure of the largest angle in a triangle with sides having lengths $3\sqrt{6}, 6\sqrt{3}, and\ 9\sqrt{2}$.
2. Find the measure of the smallest angle in a triangle with sides $7, 12\ and\ 13$.
3. In $\Delta\ RST$, $\angle R = 75^0, \angle S = 45^0\ and\ t = 3$. Find r and s.
4. In ΔABC, $cos\ A = 0.6$. Find $sin\ A$ and $tan\ A$.
5. In $\Delta\ XYZ$, $\angle X = 90^0, \angle Y = 37^0\ and\ z = 25$. Find x and y
6. From points A and B, 10 meters apart, the angles of elevation of the top of the tower are 40^0 and 60^0, respectively. Find the tower's height.
7. A ship leaves a port and sails northwest for 1 h and then northeast for 2 h. If it does not change speed, find what course the ship should take to return directly to port. Also find how long this return will take.

away so you can offer quick and suitable help to those who may need extra support or reinforcement.
To determine learners' challenges and misconceptions, ask the following questions:

- Which types of problems in this activity did you find easiest or most challenging, and why?
- How did understanding triangle properties and trigonometric ratios help you solve real-life navigation and distance problems?
- How can the skills and concepts practice in this activity be applied in everyday situations, such as travel, construction, or aviation

C.4. Additional Activities

Activity C.4: More Exercises on Triangles

Remediation
Instructions
Complete the table by providing the appropriate value/s for each angle and side an oblique triangle.

Angles and Sides	$\angle X$	$\angle Y$	$\angle Z$	x	y	z
1.		140°		15		20
2.				50	68	110
3.	50°		30°	10		
4.	45.5°	81.3°		15		
5.		20°	130°			21

Enrichment
Instructions:
Solve each problem. Write your solutions on a separate sheet of paper
1. A construction worker looks up at the top of a crane that is 60 meters tall. If the angle of elevation from the

Activity C.4: More Exercises on Triangles

This extra activity can be assigned as a problem set. This task serves remediation for learners who haven't fully mastered the lesson and offers enrichment tasks to those who are ready to advance their learning further.

Key to Corrections
Remediation

No.	∠X	∠Y	∠Z	Side x	Side y	Side z
1	140°	19°	21°	15	—	20
2	18.3°	25.3°	136.4°	50	68	110
3	50°	30°	100°	10	6.5	12.9
4	45.5°	81.3°	53.2°	15	20.8	16.9
5	30°	20°	130°	13.7	9.4	21

	worker's position to the top of the crane is 35°, how far is the worker from the base of the crane? 2. A person standing on a cliff 90 meters high spots a boat in the ocean at an angle of depression of 28°. How far is the boat from the base of the cliff? 3. A surveyor measures two sides of a triangular plot of land: one side is 120 meters, the other is 85 meters, and the angle between them is 65°. What is the length of the third side of the triangle? 4. An airplane flies 250 km on a bearing of 060°, then turns and flies 180 km on a bearing of 135°. How far is the airplane from its starting point, and what is its final bearing from the origin? 5. A ship sails 100 km on a bearing of 030°, then changes course and sails 150 km on a bearing of 120°. What is the ship's final position relative to the starting point, and what bearing should it take to return directly to port?	Enrichment 1. 85.7 m. 2. 169.3 m. 3. 114.03 m. 4. Distance from starting point: 343.8 km Final Bearing:90.4° 5. Distance from starting point: 180.3 km, Final bearing: 266.3°

VI. SUMMATIVE ASSESSMENTS	Please refer to the attachments for the summative assessment and performance task. **Answer Key: Summative Assessment** Exercise 1. 1. C 2. D 3. A 4. C 5. B 6. C 7. C 8. B 9. B 10. B Exercise 2 1. a = 24.45 $\angle B \approx 86.74°$ $\angle C \approx 51.26°$ 2. $\angle A \approx 53.18°$ $\angle B \approx 68.38°$ $\angle C \approx 58.45°$ 3. $c \approx 59.19$ $\angle A \approx 35.82°$ $\angle B \approx 24.18°$ 4. $\angle C = 37°$ Exercise 3. 1. 26.83 cm^2 2. 91.96 cm^2 3. 84 cm^2 4. 31.83 cm^2 5. 1037 miles 6. N 27.74^0 from Mactan towards Tacloban

	11. B 12. C 13. C 14. A 15. B $b = \frac{a\sin B}{\sin A} \approx 332.46$ $c = \frac{a\sin C}{\sin A} \approx 231.03$ 5. $\angle A = 65°$ $a = \frac{c\sin A}{\sin C} \approx 50.04$ $b = \frac{c\sin B}{\sin C} \approx 27.61$
VII. REFLECTION	Reflection is most effective when done regularly. Make it a habit to reflect after each lesson or weekly to track your progress and growth over time. The reflection questions are designed to help you thoughtfully evaluate your teaching, identify strengths, recognize challenges, and plan improvements to enhance student learning. The questions provided below may help in reflecting to your teaching performance. **1. What aspects of your teaching went particularly well?** • Which part of the lesson engaged students the most? • What teaching strategies or methods contributed to successful learning? • How did students respond to your instruction (e.g., participation, performance, behavior)? • What evidence shows that learning objectives were achieved? • What personal teaching strengths were evident in this session? **2. What challenges did you encounter, and how did you address them?** • What difficulties arose during lesson delivery or classroom management? • Were there moments when students struggled to understand a concept? Why? • How did you adjust your strategies or pacing to respond to these challenges? • What support or resources would have helped you handle the situation more effectively? • What will you do differently next time to overcome similar obstacles? **3 What will you do differently next time to improve student learning?** • Which teaching strategies or approaches would you modify or try in the future? • How will you better address diverse learning needs in your classroom? • Are there new resources, activities, or technologies you plan to incorporate? • What steps can you take to enhance student engagement or participation? • How will you assess whether these changes improve learning outcomes?

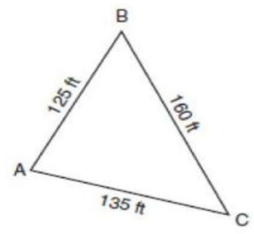
SUMMATIVE ASSESSMENT

Exercise 1. **Multiple Choices**
Direction

Read each question carefully. Choose the letter of your answer.

1. What ratio corresponds to the sine function of an acute angle in a right triangle?
   A. The ratio of the opposite and the adjacent sides
   B. The ratio of the adjacent and the hypotenuse sides
   C. The ratio of the opposite and the hypotenuse sides
   D. The ratio of the hypotenuse and the opposite sides
2. What ratio corresponds to the cosine function of an acute angle in a right triangle?
   A. The ratio of the opposite and the adjacent sides.
   B. The ratio of the opposite and the hypotenuse sides
   C. The ratio of the hypotenuse and the opposite sides
   D. The ratio of the adjacent and the hypotenuse sides
3. What ratio corresponds to the tangent function of an acute angle in a right triangle?
   A. The ratio of the opposite and the adjacent sides
   B The ratio of the adjacent and the hypotenuse sides
   C. The ratio of the opposite and the hypotenuse sides
   D. The ratio of the hypotenuse and the opposite sides
4. If angle A = 49° and side a = 10, what is the value of side c?
   A. 11.00 C. 13.25
   B. 12.50 D. 14.00
5. If side a=13 and angle B = 16°, what is the measure of the hypotenuse c?
   A. 13.00 C. 14.00
   B. 13.52 D. 15.00
6. If the hypotenuse c=16 side a=7, what is the length of side b?
   A. 12.00 C. 14.39
   B. 13.50 D. 15.00
7. If side b = 10 the hypotenuse c = 20, what is the measure of angle A?
   A. 30° C. 60°
   B. 45° D. 75°
8. In a right triangle ACB, and right angle at C, if side b=17 cm and hypotenuse c = 23cm, what is the length of side a?
   A. 14.00 cm C. 17.23 cm
   B. 15.49 cm D. 16.00 cm
9. In the right triangle ACB, and right angle at C, if the hypotenuse c = 16 and side a = 7, what is the measure of angle A?
   A. 15.24° C. 28.36°
   B. 25.94° D. 30.56°
10. If a kite string is 20 meters long and makes an angle of 40° with the ground, how high is the kite above the ground?
    A. 10.20 m C. 13.25 m
    B. 12.86 m D. 14.00 m
11. An equilateral triangle has sides whose lengths are 10 meters. What is the area of the garden?
    A. 25 m^2 C. 50 m2
    B. $25\sqrt{3}$ m^2 D. $50\sqrt{3}$ m^2

12. The accompanying diagram shows a triangular plot of land located in Moira's garden. Find the area of the plot of land, and round your answer to the nearest hundred square feet.



   A. 8177 ft^2
   B. 8190 ft^2
   C. 8200 ft^2
   D. 9000 ft^2

13. A farmer has a triangular field with sides of 240 feet, 300 feet, and 360 feet. He wants to apply fertilizer to the field. If one 40-pound bag of fertilizer covers 6,000 square feet, how many bags must he buy to cover the field?
   A. 4 bags
   B. 5 bags
   C. 6 bags
   D. 7 bags

14. A farmer is designing a triangular irrigation system. Two pipes form sides of 30 meters and 45 meters, with an angle of 60° between them. How much land will the system cover?
   A. 675 m^2
   B. 610 m^2
   C. 585 m^2
   D. 580 m^2

15. A resort in Cebu is building a triangular swimming pool. Two sides of the pool measure 18 meters and 24 meters, with an included angle of 70°.What is the area of the swimming pool?
   A. 405.00 m^2
   B. 406.50 m^2
   C. 420.00 m^2
   D. 432.00 m^2

Exercise 2. **Finding the Remaining Parts**
Direction:

In each item, find the remaining sides and angles (round in two decimal places).

1. $\angle A = 42^o, c = 25, b = 32$
2. $a = 31, b = 36,\ c = 33$
3. $\angle C = 120^o, b = 28, a = 40$
4. $\angle A = 23^o, \angle B = 120^o, a = 150$
5. $\angle B = 30^o, \angle C = 85^o, c = 55$

Exercise 3. **Problem Solving**
Direction:

Solve the following problems. Show complete solutions.

1. A triangle has sides of lengths 7 cm, 8 cm, and 9 cm. Find its area.
2. In triangle ABC, side a = 10 cm, side b = 12 cm, and angle C = 60°. Find its area.
3. A triangle has sides of 13 cm, 14 cm, and 15 cm. Calculate its area.
4. Triangle XYZ has sides x = 9 cm, y = 10 cm, and angle Z = 45°. Find the area.
5. An airplane travels in a straight line for 2.5 hours before the pilot adjusts the course, turning 20° to the left of the initial heading. The plane then continues flying in the new direction for 1 more hour. If the aircraft maintains a constant speed of 300 miles per hour, how far is it from its departure point?
6. A pilot departs from Mactan-Cebu International Airport, flying toward Dumaguete on a bearing of N60°W for a distance of 100 miles. After completing a delivery, the pilot resumes flight on a new bearing of N20°E and travels 82 miles to reach Tacloban.

Unit 7- Performance Task
Barangay Tri-Math-lon: Practical Applications of Trigonometry

Goal:
Use trigonometry and triangle-based calculations to plan community projects and solve navigation problems, demonstrating practical problem-solving skills.

Role:
You are a member of a barangay development and rescue team, acting as Surveyors, Navigators, Planners, Cartographers, or Reporters.

Audience:
Your classmates, teacher, and hypothetical community stakeholders who will review and evaluate your team's recommendations and solutions.

Situation:
Your barangay is planning projects that involve triangular spaces, such as garden plots, evacuation center roof trusses, and rice fields. Simultaneously, a rescue mission requires locating a missing vessel using bearings and distance calculations. Your team must apply trigonometric ratios, area formulas, and vector analysis to make accurate measurements, calculations, and practical recommendations.

Product/Performance:
A group presentation and report that includes:
- Height measurements of a local landmark using trigonometric ratios.
- Modeled triangular plots with calculated areas, estimated materials, and total costs.
- Navigation map showing the rescue vessel's path, final position, and bearing.
- Explanation of methods, reasoning, and real-life applications.
- Reflection on teamwork and problem-solving strategies.

Standards:
- Correct use of trigonometric ratios to measure heights and calculate distances.
- Accurate calculation of triangle areas and estimation of materials and costs.
- Correct plotting of navigation paths, distances, and bearings.
- Clear, organized, and professional presentation of results.
- Active collaboration and effective communication among team members.
- Reflection demonstrates understanding of mathematical principles and practical application.

Rubric:

Criteria	Weight	Beginning (1)	Developing (2)	Proficient (3)	Mastery (4)
1. Application of Trigonometric Ratios – Correctly uses trigonometric ratios to estimate heights or distances.	**25%**	Uses ratios incorrectly or not at all; calculations are mostly inaccurate.	Attempts to use ratios; some errors in setup or computation.	Correctly applies ratios with minor calculation errors.	Correctly applies ratios accurately and explains reasoning clearly.
2. Triangle Area Calculations & Material Estimates – Calculates	**25%**	Miscalculates areas and materials; does not interpret results.	Partially correct calculations; minor errors in estimation or interpretation.	Mostly accurate area calculations and material estimates;	Accurate calculations, clear estimation of materials/costs,

triangle areas and estimates materials and costs accurately.				interprets results appropriately.	and insightful interpretation for practical application.
3. Navigation & Bearings – Correctly calculates distances, bearings, and plots positions for rescue mission.	**20%**	Miscalculates distances/bearings; path is inaccurate.	Partially correct calculations; plot is somewhat accurate.	Correct calculations with minor errors; plot mostly accurate.	Accurate calculations and bearings; final position and path plotted clearly and correctly.
4. Presentation and Communication – Clarity, organization, visual representation, and explanation of reasoning.	**15%**	The presentation is unclear, disorganized, and lacks visuals.	Somewhat clear; visuals are incomplete; reasoning partially explained.	Clear presentation with visuals; reasoning mostly explained.	Highly organized, clear, professional visuals; reasoning explained confidently and thoroughly.
5. Teamwork & Collaboration – Effective role distribution, participation, and reflection.	**10%**	Minimal participation; poor collaboration; reflection missing or vague.	Partial participation; uneven collaboration; reflection limited.	Good participation; collaborative; reflection shows understanding.	Excellent collaboration; all roles fulfilled; reflection insightful and demonstrates learning.
6. Reflection on Practical Application – Demonstrates understanding of how trigonometric concepts apply in real-life situations.	**5%**	Reflection missing or unrelated to activity.	Reflection shows limited understanding of application.	Reflection shows good understanding and connects to real-life contexts.	Reflection is thorough, insightful, and clearly connects mathematics to real-life applications.